Multinomial Credits Analysis

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Load **survey_data:**

- survey responses from qualtrics
- includes: *5bi) How many credits of instruction in quantitative Study Design do students receive in core (compulsory) teaching across the entire undergraduate degree?* and *5ci) How many credits of undergraduate instruction in Data Analysis do students receive in core (compulsory) teaching across the entire undergraduate degree?*

```
#load survey data
survey_data <- read_rds(here("data/clean_data", "survey_data.rds"))

#look at data
survey_data %>%
  select(university_id, discipline, credits_sd, credits_da) %>%
  #sd = study design
  #da = data analysis
  glimpse()
```

```
## Rows: 103
## Columns: 4
## $ university_id <fct> u90, u90, u67, u67, u66, u67, u97, u86, u93, u12, u87, N~
## $ discipline    <fct> Bio, Psych, Bio, Med, Bio, Psych, Biomed, Biomed, Med, B~
## $ credits_sd    <ord> 20-29, 30-39, 20-29, 10-19, 1-9, 20-29, 20-29, 1-9, 1-9,~
## $ credits_da    <ord> 60-69, 40-49, 10-19, 1-9, 20-29, 50-59, 20-29, 10-19, 1-~
```

```
#table: study design credits by discipline
table(survey_data$discipline, survey_data$credits_sd)
```

```
##
##           0 1-9 10-19 20-29 30-39 40-49 50-59 60-69 70-79 80+
##   Bio     0  12     5     8     2     2     1     0     0   1
##   Biomed  3   7     4     3     0     1     0     0     0   1
##   Med     0   5     6     1     1     1     1     0     0   0
##   Psych   0   1     4    11     3     4     1     6     0   8
```

```
#table: data analysis credits by discipline
table(survey_data$discipline, survey_data$credits_da)
```

```
##
##           0 1-9 10-19 20-29 30-39 40-49 50-59 60-69 70-79 80+
##   Bio     0   6     9     8     2     3     1     1     0   1
##   Biomed  0   2    10     5     0     0     0     0     0   2
##   Med     0   8     2     4     0     0     1     0     0   0
##   Psych   0   1     1     4     8     7     2     7     0   8
```

Credits: Study Design

Models with 4 categories

recode categories

Recoding from 10 categories to 4 reduces the number of response categories with zero responses.

```
#recode categories
survey_data <-
  survey_data %>%
  mutate(
    credits_sd_recoded = fct_collapse(credits_sd,
                                      "0-19" = c("0", "1-9", "10-19"),
                                      "20-39"= c("20-29","30-39"),
                                      "40-59" = c("40-49", "50-59"),
                                      "60+" = c("60-69", "70-79", "80+")
    )
  )

#structure: ordered factor with 4 levels
glimpse(survey_data$credits_sd_recoded)
```

```
##  Ord.factor w/ 4 levels "0-19"<"20-39"<..: 2 2 2 1 1 2 2 1 1 1 ...
```

```
#table: study design credits (4 categories) by discipline
table(survey_data$discipline, survey_data$credits_sd_recoded)
```

```
##
##          0-19 20-39 40-59 60+
##   Bio      17    10     3   1
##   Biomed   14     3     1   1
##   Med      11     2     2   0
##   Psych     5    14     5  14
```

Multinomial Analysis of Study Design Credits

We now fit a multinomial model, using the prior suggested by Gelman et al. (2008) (DOI: 10.1214/08-AOAS191).

Below, we (1) identify model components that require a prior, (2) specify the prior, (3) specify contrasts (as recommended by Gelman el al. (2008)), (4) perform a prior predictive check.

```
#We obtain the default prior, and then update the prior, below.
get_prior(credits_sd_recoded ~ discipline, family = categorical(link = "logit"), data = survey_data)
```

```
##                 prior     class             coef group resp   dpar nlpar lb ub
##                (flat)         b                              mu2039
##                (flat)         b disciplineBiomed             mu2039
##                (flat)         b    disciplineMed             mu2039
##                (flat)         b  disciplinePsych             mu2039
##  student_t(3, 0, 2.5) Intercept                              mu2039
##                (flat)         b                              mu4059
##                (flat)         b disciplineBiomed             mu4059
##                (flat)         b    disciplineMed             mu4059
##                (flat)         b  disciplinePsych             mu4059
##  student_t(3, 0, 2.5) Intercept                              mu4059
##                (flat)         b                                mu60
```

```
##                 (flat)         b disciplineBiomed              mu60
##                 (flat)         b    disciplineMed              mu60
##                 (flat)         b  disciplinePsych              mu60
##  student_t(3, 0, 2.5) Intercept                                 mu60
##  tag       source
##           default
##      (vectorized)
##      (vectorized)
##      (vectorized)
##           default
##           default
##      (vectorized)
##      (vectorized)
##      (vectorized)
##           default
##           default
##      (vectorized)
##      (vectorized)
##      (vectorized)
##           default
```

```
#Our prior, following Gelman et al. (2008):
priors.studydesign <- c(
  prior(cauchy(0, 10), class = "Intercept", dpar = "mu2039"),
  prior(cauchy(0, 10), class = "Intercept", dpar = "mu4059"),
  prior(cauchy(0, 10), class = "Intercept", dpar = "mu60"),
  prior(cauchy(0, 2.5), class = "b", dpar = "mu2039"),
  prior(cauchy(0, 2.5), class = "b", dpar = "mu4059"),
  prior(cauchy(0, 2.5), class = "b", dpar = "mu60")
)

contrasts(survey_data$discipline) =contr.sum(4)
contrasts(survey_data$discipline)
```

```
##        [,1] [,2] [,3]
## Bio       1    0    0
## Biomed    0    1    0
## Med       0    0    1
## Psych    -1   -1   -1
```

```
#       [,1] [,2] [,3]
#Bio       1    0    0
#Biomed    0    1    0
#Med       0    0    1
#Psych    -1   -1   -1


#Prior predictive check:
brm.formula1.sd <- brmsformula(credits_sd_recoded ~ discipline,
                     family = categorical(link = "logit"))

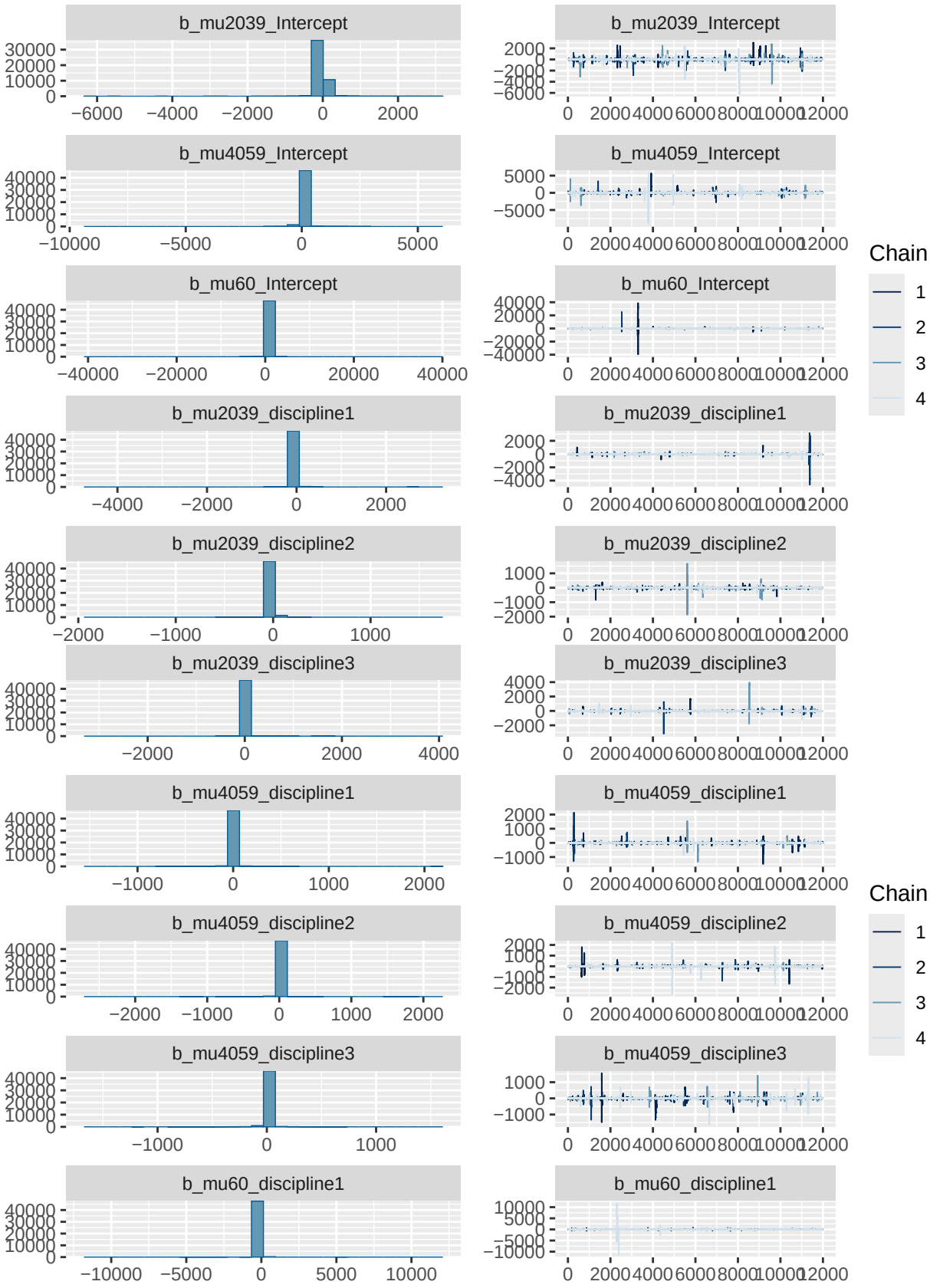
brm.mn.studydesign.priorpc <- brm(brm.formula1.sd, prior = priors.studydesign,
                                  iter = 24000,
                     control = list(adapt_delta = 0.999, stepsize = 0.001,
                                    max_treedepth = 20),
```

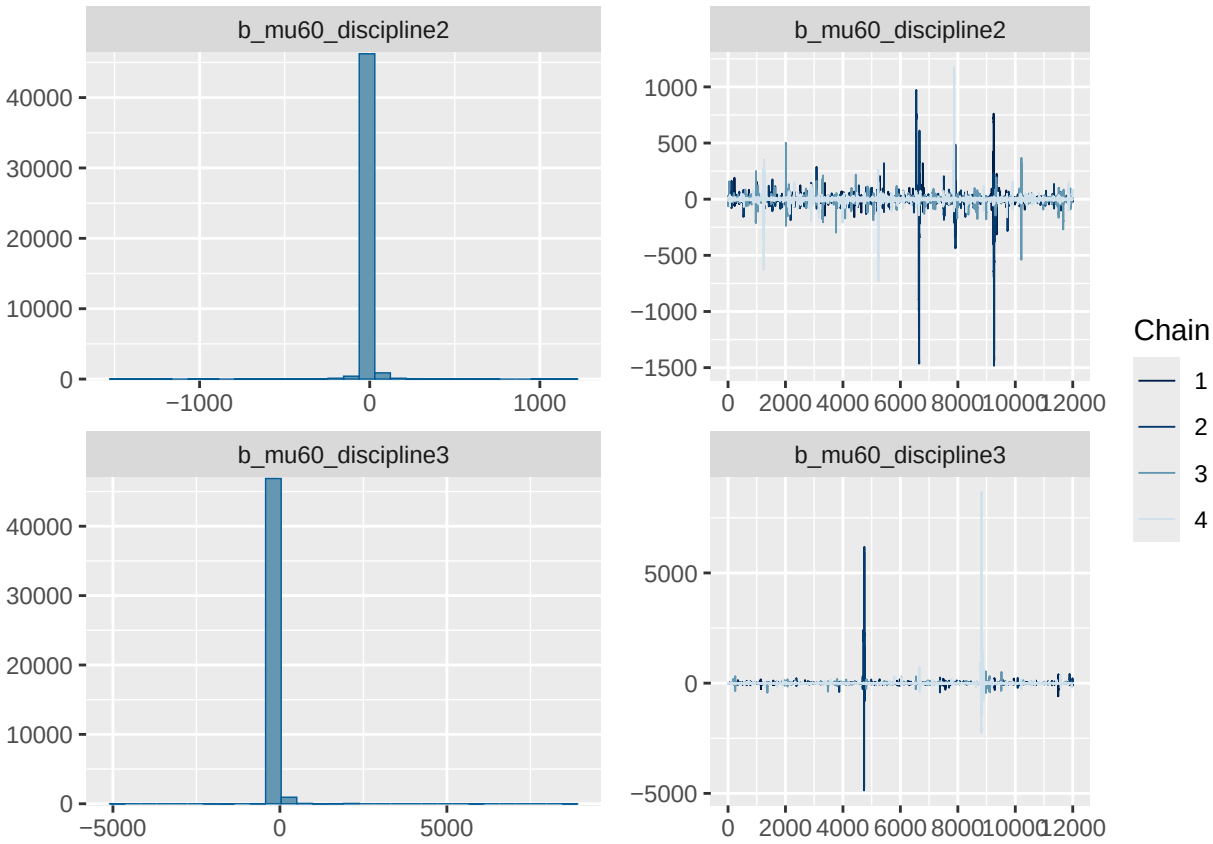
```
                                sample_prior = "only", data = survey_data,
                    file = here("models.gelman", "brm.mn.studydesign.priorpc"))

summary(brm.mn.studydesign.priorpc)
```

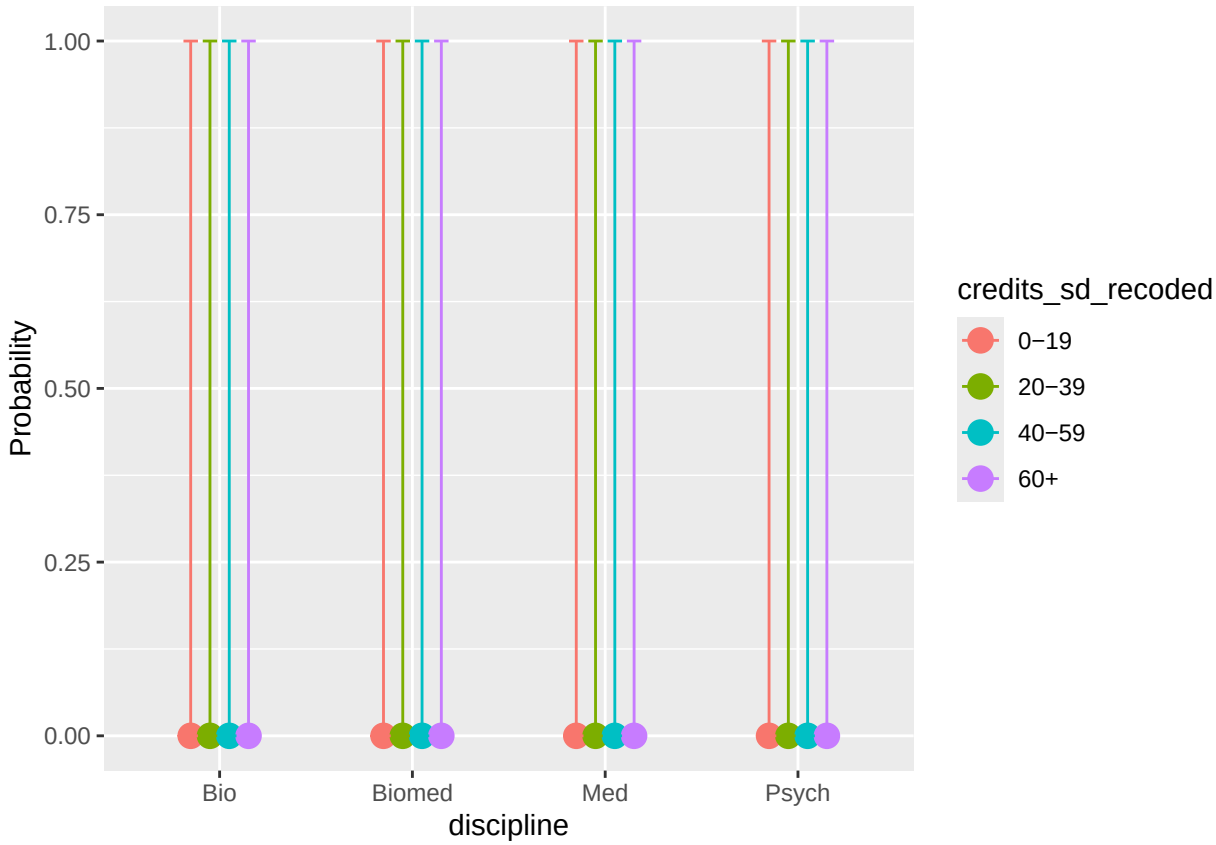
```
##  Family: categorical
##   Links: mu2039 = logit; mu4059 = logit; mu60 = logit
## Formula: credits_sd_recoded ~ discipline
##    Data: survey_data (Number of observations: 103)
##   Draws: 4 chains, each with iter = 24000; warmup = 12000; thin = 1;
##          total post-warmup draws = 48000
##
## Regression Coefficients:
##                    Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## mu2039_Intercept      -4.84    255.87  -151.65   147.84 1.00    13664     3738
## mu4059_Intercept       2.77    232.60  -131.69   152.45 1.00    14559     4266
## mu60_Intercept        13.33    639.92  -121.72   131.63 1.00    14789     3431
## mu2039_discipline1     1.01     92.71   -27.67    28.68 1.00    26815     7851
## mu2039_discipline2    -0.61     40.01   -30.66    31.48 1.00    19540     5537
## mu2039_discipline3     1.85     90.55   -32.35    32.68 1.00    25229     5866
## mu4059_discipline1     0.79     64.13   -30.41    31.60 1.00    17586     4612
## mu4059_discipline2    -0.12     77.55   -28.89    27.48 1.00    26491     6513
## mu4059_discipline3    -1.25     62.29   -38.41    33.41 1.00    18456     4138
## mu60_discipline1      -0.36    251.00   -34.52    35.32 1.00    21305     5653
## mu60_discipline2      -0.78     50.56   -31.66    27.87 1.00    17911     4539
## mu60_discipline3       3.97    153.67   -29.64    34.62 1.00    21920     4178
##
## Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
## and Tail_ESS are effective sample size measures, and Rhat is the potential
## scale reduction factor on split chains (at convergence, Rhat = 1).
```

```
plot(brm.mn.studydesign.priorpc)
```


b_mu2039_Intercept
b_mu4059_Intercept
b_mu60_Intercept
b_mu2039_discipline1
b_mu2039_discipline2
b_mu2039_discipline3
b_mu4059_discipline1
b_mu4059_discipline2
b_mu4059_discipline3
b_mu60_discipline1
Chain
1
2
3
4




```
#Prior predictive check outout:
conditional_effects(brm.mn.studydesign.priorpc, categorical = TRUE)
```



Output from `summary(brm.mn.studydesign.priorpc)` and `plot(brm.mn.studydesign.priorpc)` both indicate that estimates are centered on zero (the prior has little influence on estimates).

NOTE: exploratory analyses (not shown) indicate that stronger priors yield estimates that approach 25% for each discipline (with narrower credible intervals) in prior predictive checks (`conditional_effects(brm.mn.studydesign.prio categorical = TRUE)`).

Now we run the multinomial model with our specified prior, and check for appropriate model behaviour:

```
#Run the model with the Gelman prior:
brm.mn.studydesign <- brm(credits_sd_recoded ~ discipline,
                      family = categorical(link = "logit"),
                      prior = priors.studydesign,
                      iter = 6000,
                      control = list(adapt_delta = 0.999, stepsize = 0.001,
                                     max_treedepth = 20),
                      data = survey_data,
                      file = here("models.gelman", "brm.mn.studydesign"))

#Examine prior that we specified:
prior_summary(brm.mn.studydesign)
```

```
##           prior     class        coef group resp   dpar nlpar lb ub tag
##  cauchy(0, 2.5)         b                         mu2039
##  cauchy(0, 2.5)         b disciplineBio           mu2039
##  cauchy(0, 2.5)         b disciplineBiomed        mu2039
##  cauchy(0, 2.5)         b disciplineMed           mu2039
##  cauchy(0, 2.5)         b disciplinePsych         mu2039
##  cauchy(0, 2.5)         b                         mu4059
##  cauchy(0, 2.5)         b disciplineBio           mu4059
```

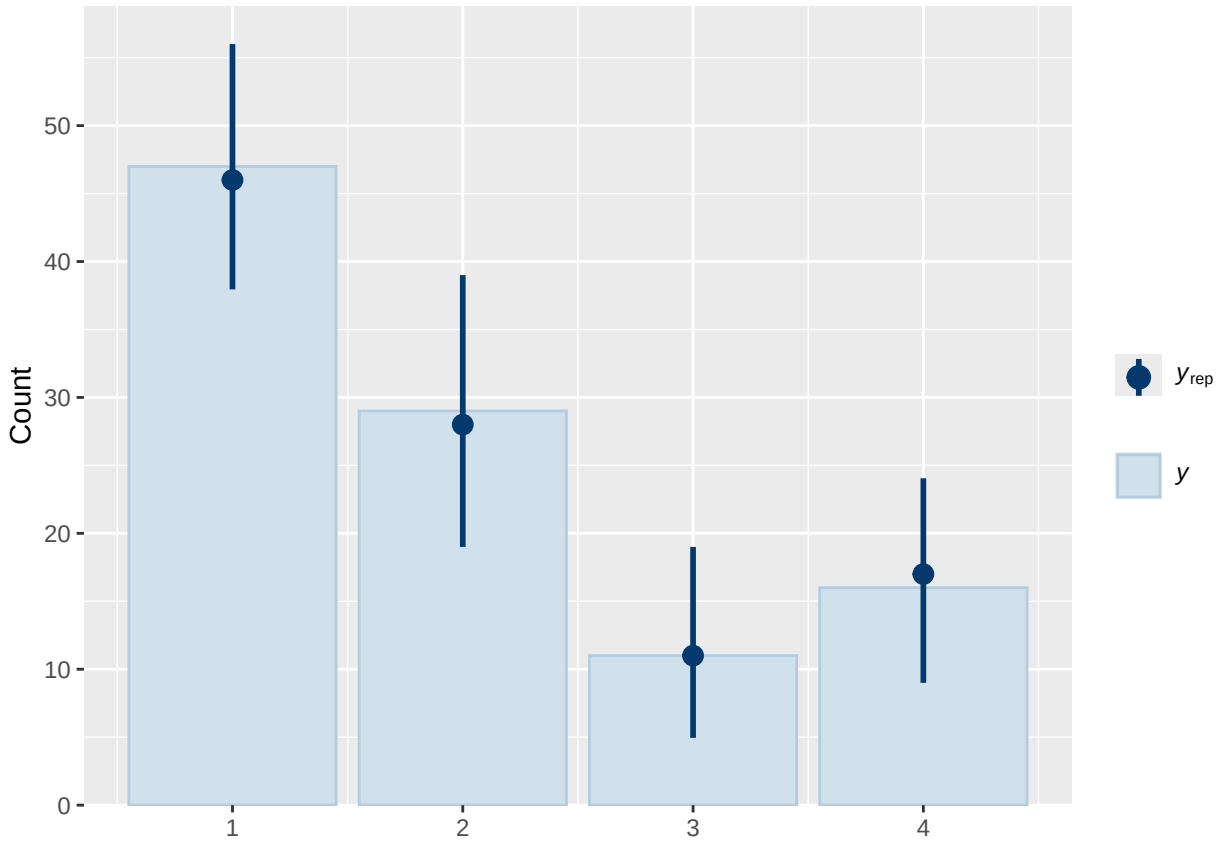
```
##  cauchy(0, 2.5)         b discipline2            mu4059
##  cauchy(0, 2.5)         b discipline3            mu4059
##  cauchy(0, 2.5)         b                          mu60
##  cauchy(0, 2.5)         b discipline1              mu60
##  cauchy(0, 2.5)         b discipline2              mu60
##  cauchy(0, 2.5)         b discipline3              mu60
##   cauchy(0, 10) Intercept                        mu2039
##   cauchy(0, 10) Intercept                        mu4059
##   cauchy(0, 10) Intercept                          mu60
##        source
##          user
##  (vectorized)
##  (vectorized)
##  (vectorized)
##          user
##  (vectorized)
##  (vectorized)
##  (vectorized)
##          user
##  (vectorized)
##  (vectorized)
##  (vectorized)
##          user
##          user
##          user
```

```
#Check convergence and coverage:
#We want Rhat to be less than 1.01
#We want _ESS to be 100 times the number of chains we run
summary(brm.mn.studydesign)
```

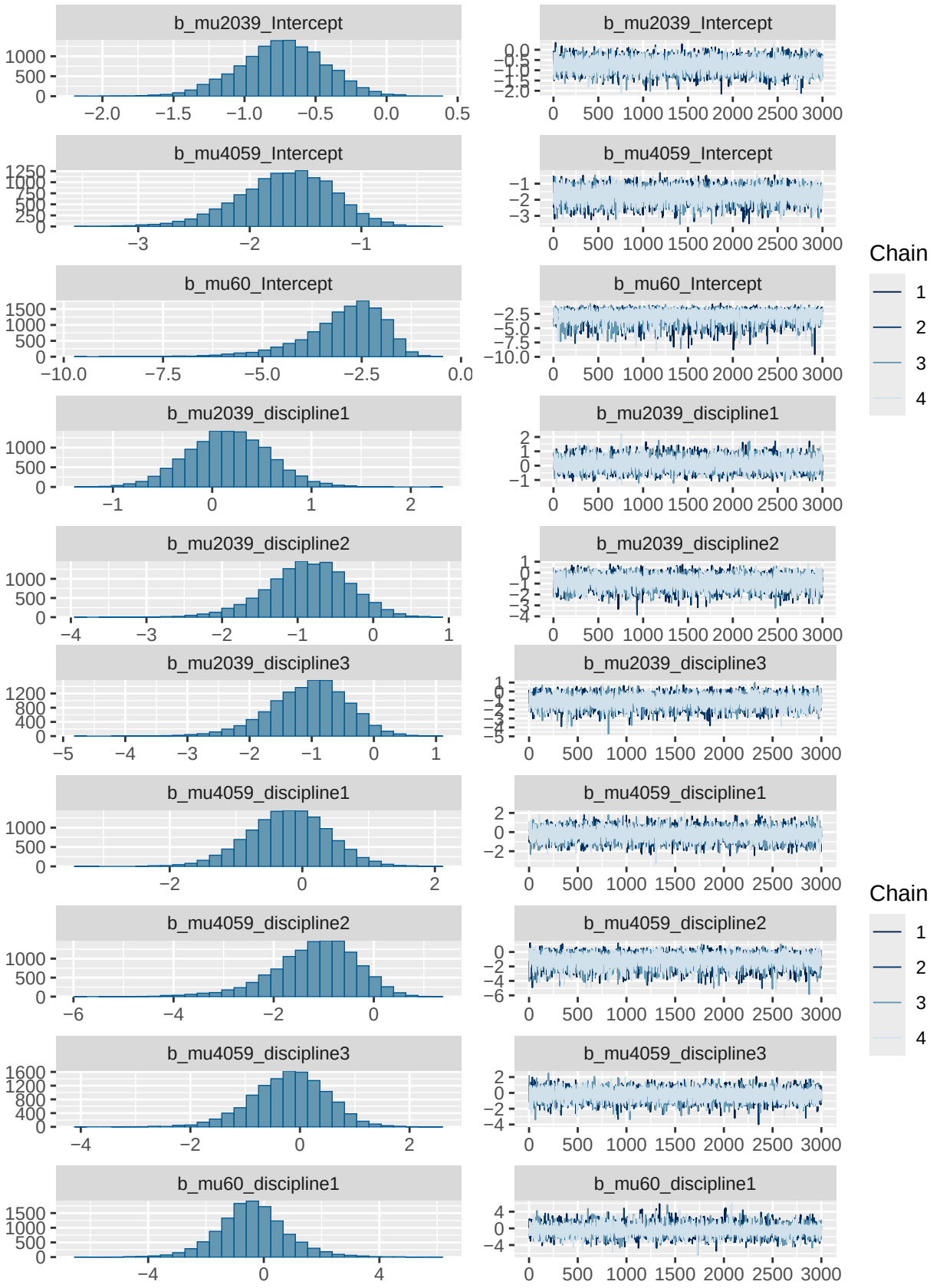
```
##  Family: categorical
##   Links: mu2039 = logit; mu4059 = logit; mu60 = logit
## Formula: credits_sd_recoded ~ discipline
##    Data: survey_data (Number of observations: 103)
##   Draws: 4 chains, each with iter = 6000; warmup = 3000; thin = 1;
##          total post-warmup draws = 12000
##
## Regression Coefficients:
##                    Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## mu2039_Intercept      -0.74      0.31    -1.36    -0.17 1.00     7916     7898
## mu4059_Intercept      -1.68      0.43    -2.58    -0.91 1.00     7945     7431
## mu60_Intercept        -2.96      1.04    -5.55    -1.47 1.00     3824     2874
## mu2039_discipline1     0.16      0.41    -0.63     0.97 1.00     8007     7540
## mu2039_discipline2    -0.87      0.55    -2.01     0.13 1.00     8924     7569
## mu2039_discipline3    -1.05      0.63    -2.42     0.06 1.00     8481     6499
## mu4059_discipline1    -0.21      0.60    -1.42     0.94 1.00     8011     7190
## mu4059_discipline2    -1.19      0.84    -3.15     0.22 1.00     7210     5681
## mu4059_discipline3    -0.20      0.68    -1.61     1.04 1.00     8195     7248
## mu60_discipline1      -0.38      1.21    -2.73     2.19 1.00     4630     4221
## mu60_discipline2      -0.21      1.23    -2.60     2.39 1.00     4908     3874
## mu60_discipline3      -3.38      2.60   -10.02     0.01 1.00     3931     2792
##
## Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
## and Tail_ESS are effective sample size measures, and Rhat is the potential
```

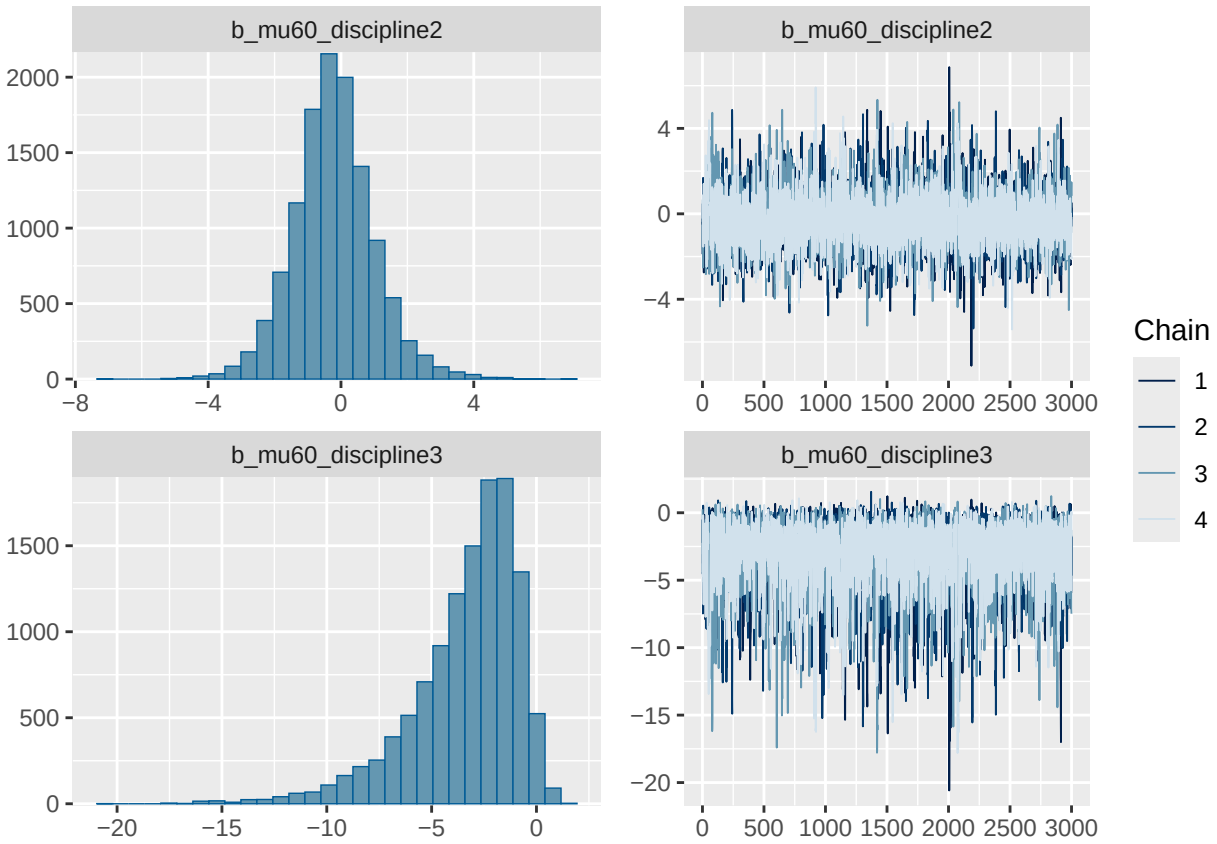
```
## scale reduction factor on split chains (at convergence, Rhat = 1).
```

```
#Posterior predictive check:
pp_check(brm.mn.studydesign, type = "bars", ndraws = 100)
```




```
#Check traces
plot(brm.mn.studydesign)
```


b_mu2039_Intercept
b_mu4059_Intercept
b_mu60_Intercept
b_mu2039_discipline1
b_mu2039_discipline2
b_mu2039_discipline3
b_mu4059_discipline1
b_mu4059_discipline2
b_mu4059_discipline3
b_mu60_discipline1
Chain
1
2
3
4



The summary (`summary(brm.mn.studydesign)`) indicates that the model converged well (based on Rhat) and has high coverage for all estimates.

The posterior predictive check output (`pp_check(brm.mn.studydesign, type = "bars", ndraws = 100)`) indicates sufficient model fit.

The traces and distributions of estimates on the latent scale (`plot(brm.mn.studydesign)`) also indicate good model behaviour.

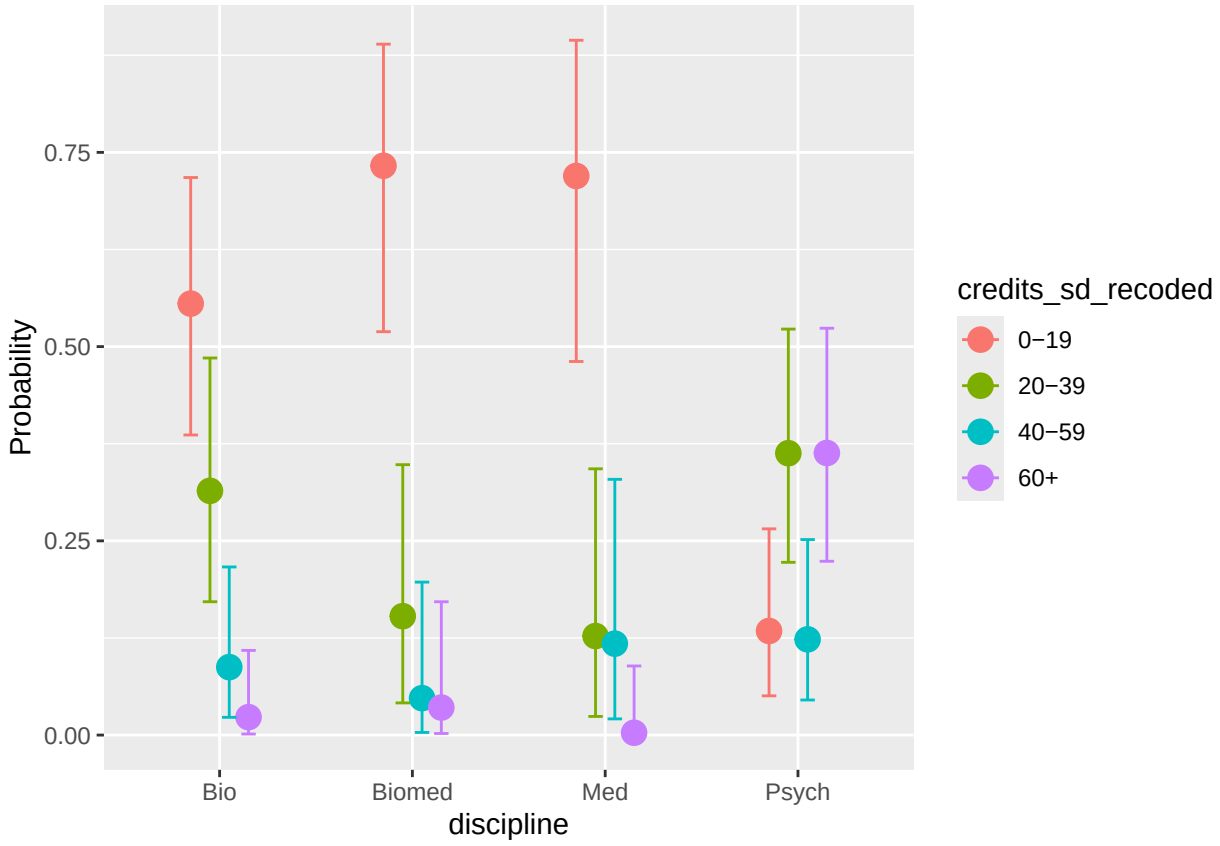
Now we obtain the model estimates:

```
eff_brm.mn.studydesign <- conditional_effects(brm.mn.studydesign, categorical = TRUE)
eff_brm.mn.studydesign[[1]]
```

```
##    discipline credits_sd_recoded cond__ cats__ effect1__ effect2__  estimate__
## 1         Bio               0-19      1   0-19       Bio      0-19 0.555342014
## 2      Biomed               0-19      1   0-19    Biomed      0-19 0.732768934
## 3         Med               0-19      1   0-19       Med      0-19 0.719664644
## 4       Psych               0-19      1   0-19     Psych      0-19 0.134010013
## 5         Bio               0-19      1  20-39       Bio     20-39 0.314358612
## 6      Biomed               0-19      1  20-39    Biomed     20-39 0.153032677
## 7         Med               0-19      1  20-39       Med     20-39 0.127485742
## 8       Psych               0-19      1  20-39     Psych     20-39 0.362822679
## 9         Bio               0-19      1  40-59       Bio     40-59 0.087468550
## 10     Biomed               0-19      1  40-59    Biomed     40-59 0.047517907
## 11        Med               0-19      1  40-59       Med     40-59 0.117675262
## 12      Psych               0-19      1  40-59     Psych     40-59 0.123201783
## 13        Bio               0-19      1    60+       Bio       60+ 0.023092594
## 14     Biomed               0-19      1    60+    Biomed       60+ 0.035449771
```

```
## 15        Med               0-19      1    60+       Med       60+ 0.003005267
## 16      Psych               0-19      1    60+     Psych       60+ 0.363082635
##             se__      lower__    upper__
## 1   0.087705090 3.862343e-01 0.71762919
## 2   0.095859533 5.191941e-01 0.88930248
## 3   0.111495515 4.808296e-01 0.89431219
## 4   0.054437011 5.049724e-02 0.26544459
## 5   0.082043889 1.716382e-01 0.48534823
## 6   0.077810928 4.139571e-02 0.34801698
## 7   0.079328303 2.396683e-02 0.34267415
## 8   0.078386508 2.223363e-01 0.52254096
## 9   0.048391200 2.284172e-02 0.21636718
## 10 0.041459080 3.354536e-03 0.19689781
## 11 0.075889620 2.082400e-02 0.32915124
## 12 0.051495433 4.513418e-02 0.25166701
## 13 0.022439187 1.335813e-03 0.10898290
## 14 0.034468601 2.035059e-03 0.17158441
## 15 0.004433324 1.528118e-07 0.08895391
## 16 0.077365059 2.236843e-01 0.52365840
```

```
print(eff_brm.mn.studydesign)
```



Plot

```
plot_colour <- "skyblue4"
```

```
plot_theme <- theme_bw() +
  theme(
    plot.title = element_text(size = 18),
    axis.title.x = element_text(size = 18),
    axis.text.x = element_text(size= 16),
    axis.title.y = element_text(size= 24),
    axis.text.y = element_markdown(size = 16, hjust = 1),
    axis.ticks.y = element_blank(),
    strip.text = element_text(size = 20),
    strip.background = element_blank(),
    panel.border = element_blank(),
    panel.grid.major = element_blank(),
    panel.grid.minor = element_blank(),
    legend.position = "none")
```

```
study_design_estimates <- eff_brm.mn.studydesign[[1]] %>%
  clean_names() %>%
  rename(
    credits_sd = cats,
    probability_estimate = estimate) %>%
  select(discipline, credits_sd, probability_estimate, lower, upper)

study_design_estimates
```

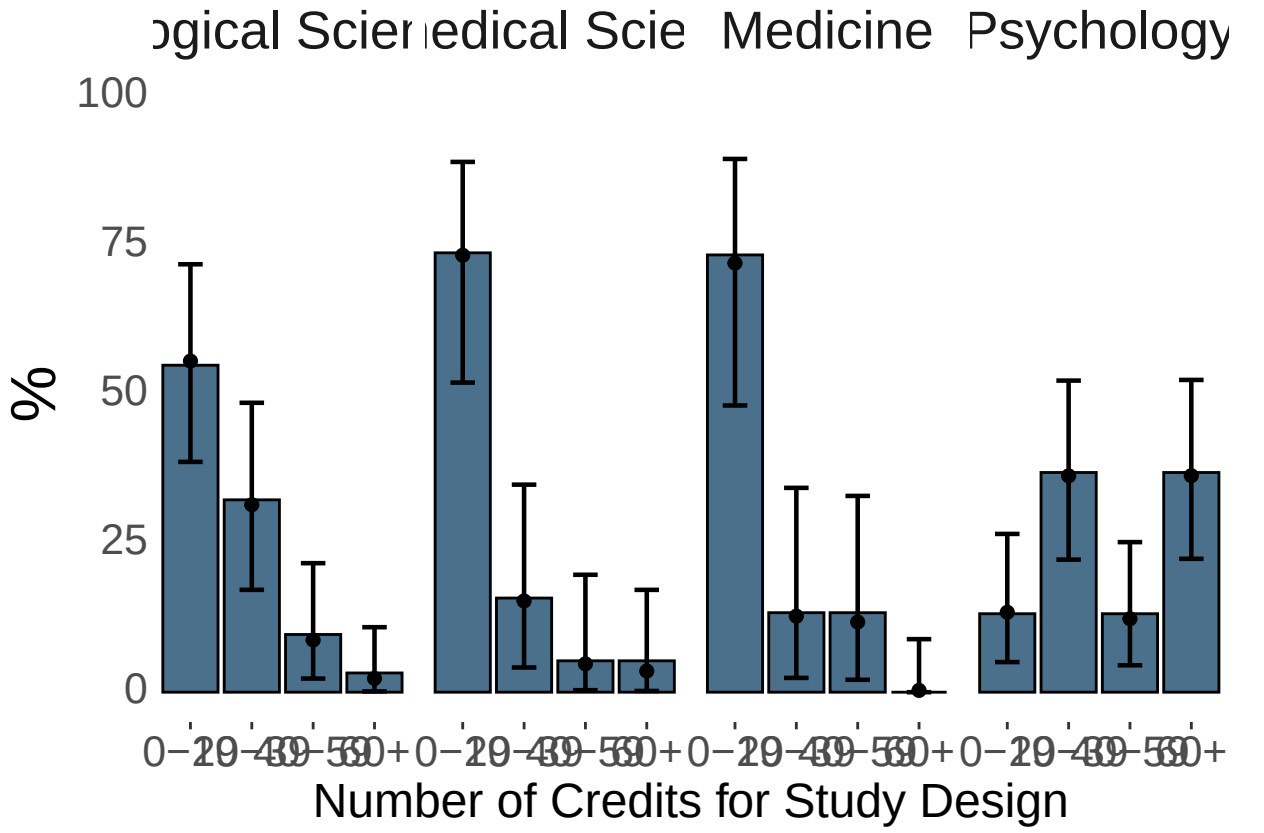
```
##    discipline credits_sd probability_estimate        lower      upper
## 1         Bio       0-19          0.555342014 3.862343e-01 0.71762919
## 2      Biomed       0-19          0.732768934 5.191941e-01 0.88930248
## 3         Med       0-19          0.719664644 4.808296e-01 0.89431219
## 4       Psych       0-19          0.134010013 5.049724e-02 0.26544459
## 5         Bio      20-39          0.314358612 1.716382e-01 0.48534823
## 6      Biomed      20-39          0.153032677 4.139571e-02 0.34801698
## 7         Med      20-39          0.127485742 2.396683e-02 0.34267415
## 8       Psych      20-39          0.362822679 2.223363e-01 0.52254096
## 9         Bio      40-59          0.087468550 2.284172e-02 0.21636718
## 10     Biomed      40-59          0.047517907 3.354536e-03 0.19689781
## 11        Med      40-59          0.117675262 2.082400e-02 0.32915124
## 12      Psych      40-59          0.123201783 4.513418e-02 0.25166701
## 13        Bio        60+          0.023092594 1.335813e-03 0.10898290
## 14     Biomed        60+          0.035449771 2.035059e-03 0.17158441
## 15        Med        60+          0.003005267 1.528118e-07 0.08895391
## 16      Psych        60+          0.363082635 2.236843e-01 0.52365840
```

```
survey_table <- table(survey_data$credits_sd_recoded, survey_data$discipline)
survey_table <- prop.table(survey_table, margin = 2)
survey_table <- as.data.frame(survey_table)
survey_ggplot <- survey_table %>%
  rename(
    credits_sd = Var1,
    discipline = Var2) %>%
  mutate(percentage = 100*Freq)
```

```
study_design_plot <- ggplot() +
  geom_col(
    data = survey_ggplot,
```

```
    aes(x = credits_sd, y = percentage),
    color = "black",
    fill = plot_colour) +
  geom_errorbar(
    data = study_design_estimates,
    aes(x = credits_sd, ymin = 100*lower, ymax = 100*upper),
    width = .4,
    linewidth = .8) +
  geom_point(
    data = study_design_estimates,
    aes(x = credits_sd, y = 100*probability_estimate),
    size = 2) +
  scale_x_discrete(drop = FALSE) +
  scale_y_continuous(
    "%",
    limits = c(0, 100),
    labels = c(0, 25, 50, 75, 100)) +
  facet_wrap(
    ~ factor(
      discipline,
      labels = c("Biological Sciences", "Biomedical Sciences", "Medicine", "Psychology")),
    nrow = 1) +
  xlab("Number of Credits for Study Design") +
  plot_theme


study_design_plot
```



```
ggsave(
  plot = study_design_plot,
  here("figures", "credits_sd_plot_gelman.jpg"),
  device = "jpeg",
  width = 1280,
  height = 637,
  units = "px",
  dpi = 100)
```

Perform Sensitivity Analysis:

```
#Sensitivity analysis:  determines the extent to which the prior affects the results
#(a qualitative test, not a quantitative test)

brm.mn_fixed <- tidy(brm.mn.studydesign, effects = "fixed", conf.level = 0.95, fix.intercept = FALSE) %>%
  mutate(
    ci_width = abs(conf.low - conf.high)
  )
```

```
## Warning in tidy.brmsfit(brm.mn.studydesign, effects = "fixed", conf.level =
## 0.95, : some parameter names contain underscores: term naming may be
## unreliable!
```

```
brm.mn_fixed
```

```
## # A tibble: 12 x 8
##    effect component term          estimate std.error conf.low conf.high ci_width
##    <chr>  <chr>     <chr>            <dbl>     <dbl>    <dbl>     <dbl>    <dbl>
##  1 fixed  cond      mu2039_Inter~   -0.738     0.305   -1.36   -0.168       1.19
##  2 fixed  cond      mu4059_Inter~   -1.68      0.425   -2.58   -0.908       1.67
##  3 fixed  cond      mu60_Interce~   -2.96      1.04    -5.55   -1.47        4.07
##  4 fixed  cond      mu2039_disci~    0.162     0.410   -0.633   0.973       1.61
##  5 fixed  cond      mu2039_disci~   -0.866     0.545   -2.01    0.129       2.14
##  6 fixed  cond      mu2039_disci~   -1.05      0.629   -2.42    0.0597      2.48
##  7 fixed  cond      mu4059_disci~   -0.212     0.603   -1.42    0.939       2.36
##  8 fixed  cond      mu4059_disci~   -1.19      0.844   -3.15    0.224       3.37
##  9 fixed  cond      mu4059_disci~   -0.198     0.683   -1.61    1.04        2.65
## 10 fixed  cond      mu60_discipl~   -0.382     1.21    -2.73    2.19        4.93
## 11 fixed  cond      mu60_discipl~   -0.214     1.23    -2.60    2.39        4.99
## 12 fixed  cond      mu60_discipl~   -3.38      2.60   -10.0     0.00867    10.0
```

```
brm.mn_fixed_reduced <- brm.mn_fixed %>%
  select(
    effect, component, term, estimate
  )
brm.mn_fixed_reduced
```

```
## # A tibble: 12 x 4
##    effect component term                estimate
##    <chr>  <chr>     <chr>                  <dbl>
##  1 fixed  cond      mu2039_Intercept      -0.738
##  2 fixed  cond      mu4059_Intercept      -1.68
##  3 fixed  cond      mu60_Intercept        -2.96
##  4 fixed  cond      mu2039_discipline1     0.162
##  5 fixed  cond      mu2039_discipline2    -0.866
##  6 fixed  cond      mu2039_discipline3    -1.05
```

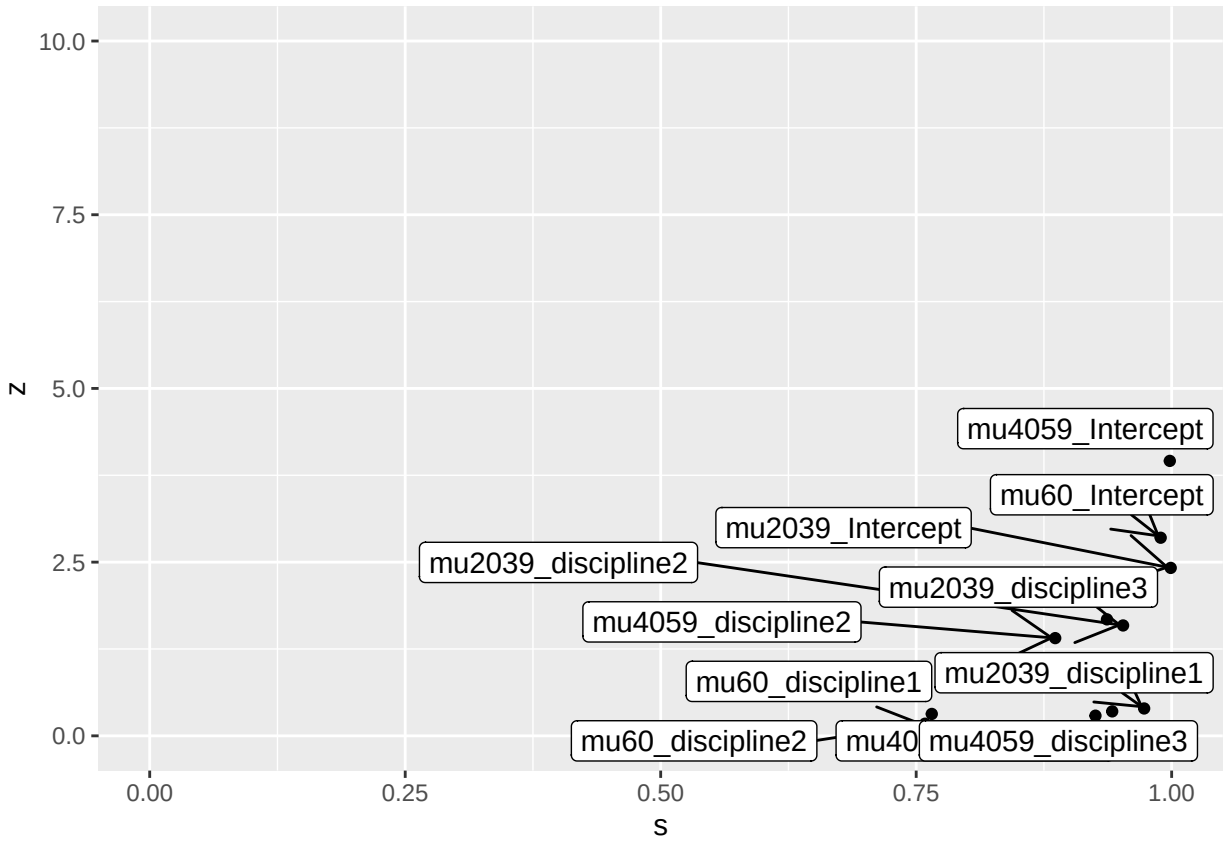
```
##  7 fixed  cond      mu4059_discipline1   -0.212
##  8 fixed  cond      mu4059_discipline2   -1.19
##  9 fixed  cond      mu4059_discipline3   -0.198
## 10 fixed  cond      mu60_discipline1     -0.382
## 11 fixed  cond      mu60_discipline2     -0.214
## 12 fixed  cond      mu60_discipline3     -3.38
```

```r
#We have 12 fixed effects; the first 3 are intercepts, and the last 9 are coefficients (b)
#Also, the 3 Intercepts are listed first in the output of brm.mn_fixed,
#and the Intercepts had priors with sd = 10
#The remaining 9 fixed effects are for b, which had sd = 2.5 for the prior.
#Hence, we set up the sensitivity analysis like this:
brm.mn.fixed <- brm.mn_fixed %>%
  mutate(
    theta = c(0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0),
    sigma_prior = c(10, 10, 10, 2.5, 2.5, 2.5, 2.5, 2.5, 2.5, 2.5, 2.5, 2.5),
    z = abs((estimate - theta) / std.error), # it's called here std.error but is the standard deviation
    s = 1 - (std.error^2 / sigma_prior^2)
  )

  ggplot(brm.mn.fixed, aes(s, z, label = term)) +
  geom_point() +
  geom_label_repel(arrow = arrow()) +
  xlim(0, 1) + ylim(0, 10)
```

```
## Warning: Removed 1 row containing missing values or values outside the scale range
## (`geom_point()`).
```

```
## Warning: Removed 1 row containing missing values or values outside the scale range
## (`geom_label_repel()`).
```




```
brm.mn.fixed$s
```

```
##   b_mu2039_Intercept   b_mu4059_Intercept     b_mu60_Intercept
##           0.99906676           0.99819330           0.98925877
## b_mu2039_discipline1 b_mu2039_discipline2 b_mu2039_discipline3
##           0.97307805           0.95245874           0.93671204
## b_mu4059_discipline1 b_mu4059_discipline2 b_mu4059_discipline3
##           0.94183071           0.88605452           0.92543784
##   b_mu60_discipline1   b_mu60_discipline2   b_mu60_discipline3
##           0.76526213           0.75865362          -0.08363055
```

```
brm.mn.fixed$z
```

```
##   b_mu2039_Intercept   b_mu4059_Intercept     b_mu60_Intercept
##            2.4171012            3.9570861            2.8512912
## b_mu2039_discipline1 b_mu2039_discipline2 b_mu2039_discipline3
##            0.3942104            1.5888421            1.6762526
## b_mu4059_discipline1 b_mu4059_discipline2 b_mu4059_discipline3
##            0.3520711            1.4066050            0.2903989
##   b_mu60_discipline1   b_mu60_discipline2   b_mu60_discipline3
##            0.3154975            0.1745188            1.2995076
```

Credits: Data Analysis

Models with 4 categories

recode categories

```
#recode categories
survey_data <-
  survey_data %>%
  mutate(
    credits_da_recoded = fct_collapse(credits_da,
                                      "0-19" = c("0", "1-9", "10-19"),
                                      "20-39"= c("20-29","30-39"),
                                      "40-59" = c("40-49", "50-59"),
                                      "60+" = c("60-69", "70-79", "80+")
    )
  )

#structure: ordered factor with 4 levels
glimpse(survey_data$credits_da_recoded)
```

```
##  Ord.factor w/ 4 levels "0-19"<"20-39"<..: 4 3 1 1 2 3 2 1 1 1 ...
```

```
#table: data analysis credits (4 categories) by discipline
table(survey_data$discipline, survey_data$credits_da_recoded)
```

```
##
##          0-19 20-39 40-59 60+
##   Bio      15    10     4   2
##   Biomed   12     5     0   2
##   Med      10     4     1   0
##   Psych     2    12     9  15
```

Multinomial Analysis of Data Analysis Credits

We now fit a multinomial model, using the prior suggested by Gelman et al. (2008) (DOI: 10.1214/08-AOAS191)

Below, we (1) identify model components that require a prior, (2) specify the prior, (3) specify contrasts (as recommended by Gelman el al. (2008)), (4) perform a prior predictive check.

```
#We obtain the default prior, and then update the prior, below.
get_prior(credits_da_recoded ~ discipline, family = categorical(link = "logit"),
          data = survey_data)
```

```
##                 prior     class        coef group resp   dpar nlpar lb ub tag
##               (flat)         b                        mu2039
##               (flat)         b disciplineBiomed       mu2039
##               (flat)         b disciplineMed          mu2039
##               (flat)         b disciplinePsych        mu2039
##  student_t(3, 0, 2.5) Intercept                        mu2039
##               (flat)         b                        mu4059
##               (flat)         b disciplineBiomed       mu4059
##               (flat)         b disciplineMed          mu4059
##               (flat)         b disciplinePsych        mu4059
##  student_t(3, 0, 2.5) Intercept                        mu4059
##               (flat)         b                          mu60
##               (flat)         b disciplineBiomed         mu60
```

```
##                (flat)         b discipline2              mu60
##                (flat)         b discipline3              mu60
##  student_t(3, 0, 2.5) Intercept                           mu60
##        source
##       default
##  (vectorized)
##  (vectorized)
##  (vectorized)
##       default
##       default
##  (vectorized)
##  (vectorized)
##  (vectorized)
##       default
##       default
##  (vectorized)
##  (vectorized)
##  (vectorized)
##       default
```

```
#Our prior, following Gelman et al. (2008):
priors.dataanalysis <- c(
  prior(cauchy(0, 10), class = "Intercept", dpar = "mu2039"),
  prior(cauchy(0, 10), class = "Intercept", dpar = "mu4059"),
  prior(cauchy(0, 10), class = "Intercept", dpar = "mu60"),
  prior(cauchy(0, 2.5), class = "b", dpar = "mu2039"),
  prior(cauchy(0, 2.5), class = "b", dpar = "mu4059"),
  prior(cauchy(0, 2.5), class = "b", dpar = "mu60")
)

contrasts(survey_data$discipline) =contr.sum(4)
contrasts(survey_data$discipline)
```

```
##        [,1] [,2] [,3]
## Bio       1    0    0
## Biomed    0    1    0
## Med       0    0    1
## Psych    -1   -1   -1
```

```
#       [,1] [,2] [,3]
#Bio       1    0    0
#Biomed    0    1    0
#Med       0    0    1
#Psych    -1   -1   -1

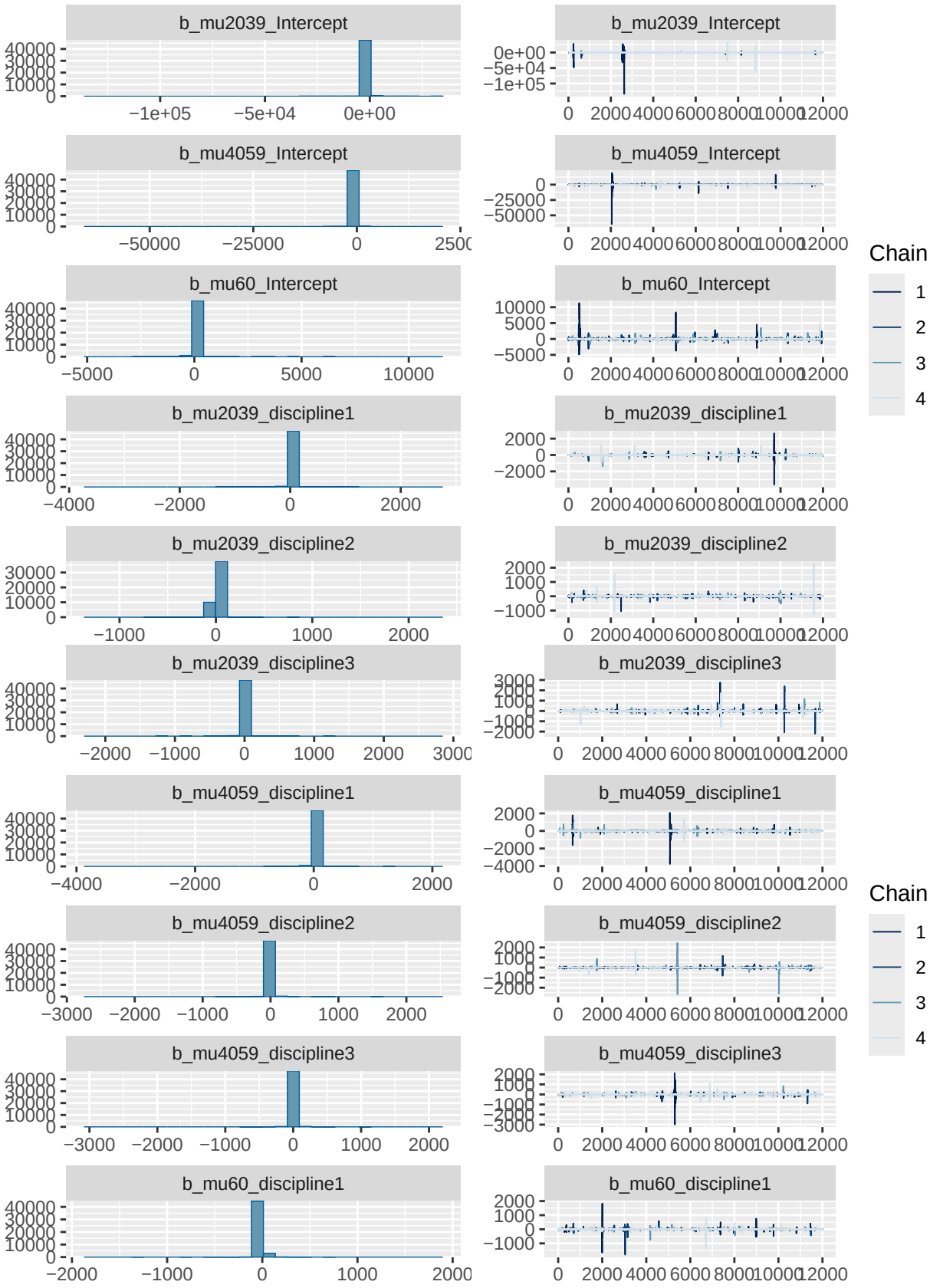
#Prior predictive check:
brm.formula1.da <- brmsformula(credits_da_recoded ~ discipline,
                     family = categorical(link = "logit"))

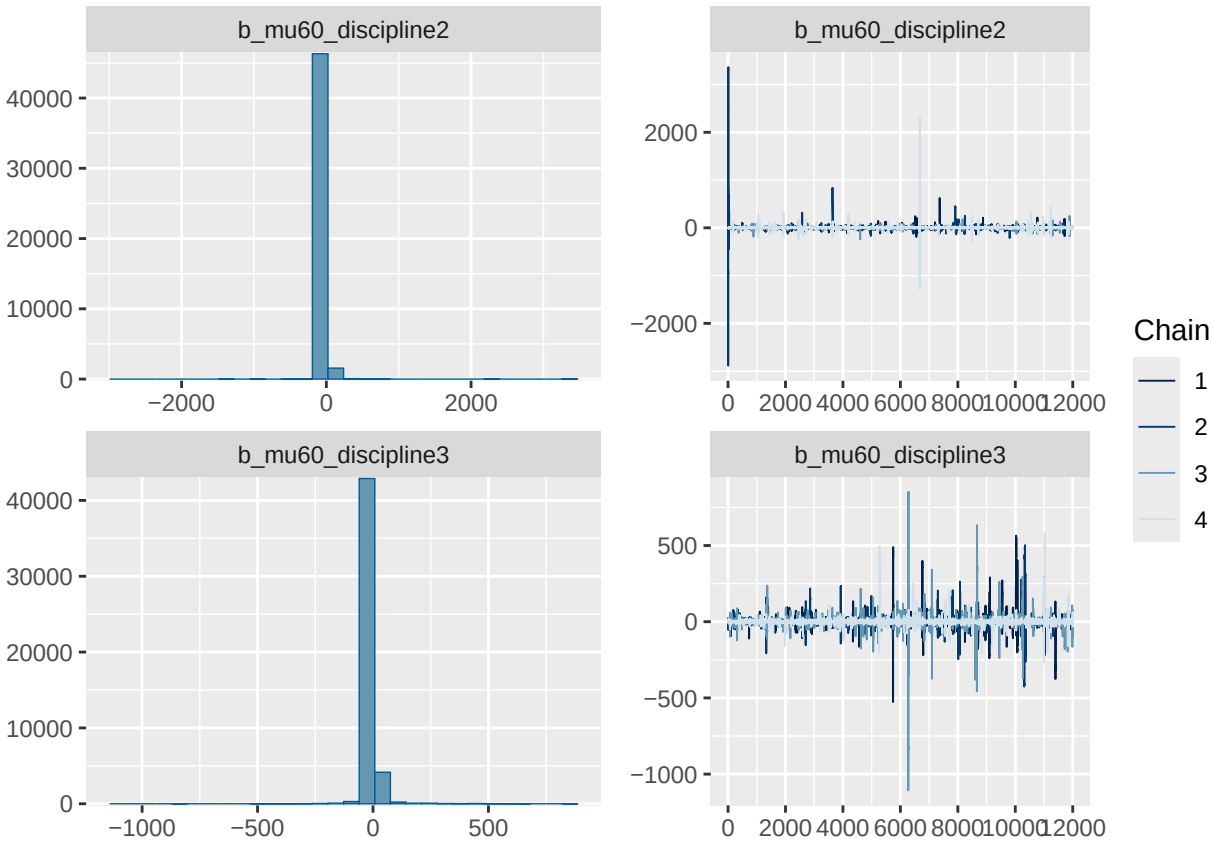
brm.mn.dataanalysis.priorpc <- brm(brm.formula1.da, prior = priors.dataanalysis,
                                   iter = 24000,
                     control = list(adapt_delta = 0.999, stepsize = 0.001,
                                    max_treedepth = 20),
                                   sample_prior = "only", data = survey_data,
                     file = here("models.gelman", "brm.mn.dataanalysis.priorpc"))
```

```r
summary(brm.mn.dataanalysis.priorpc)
```

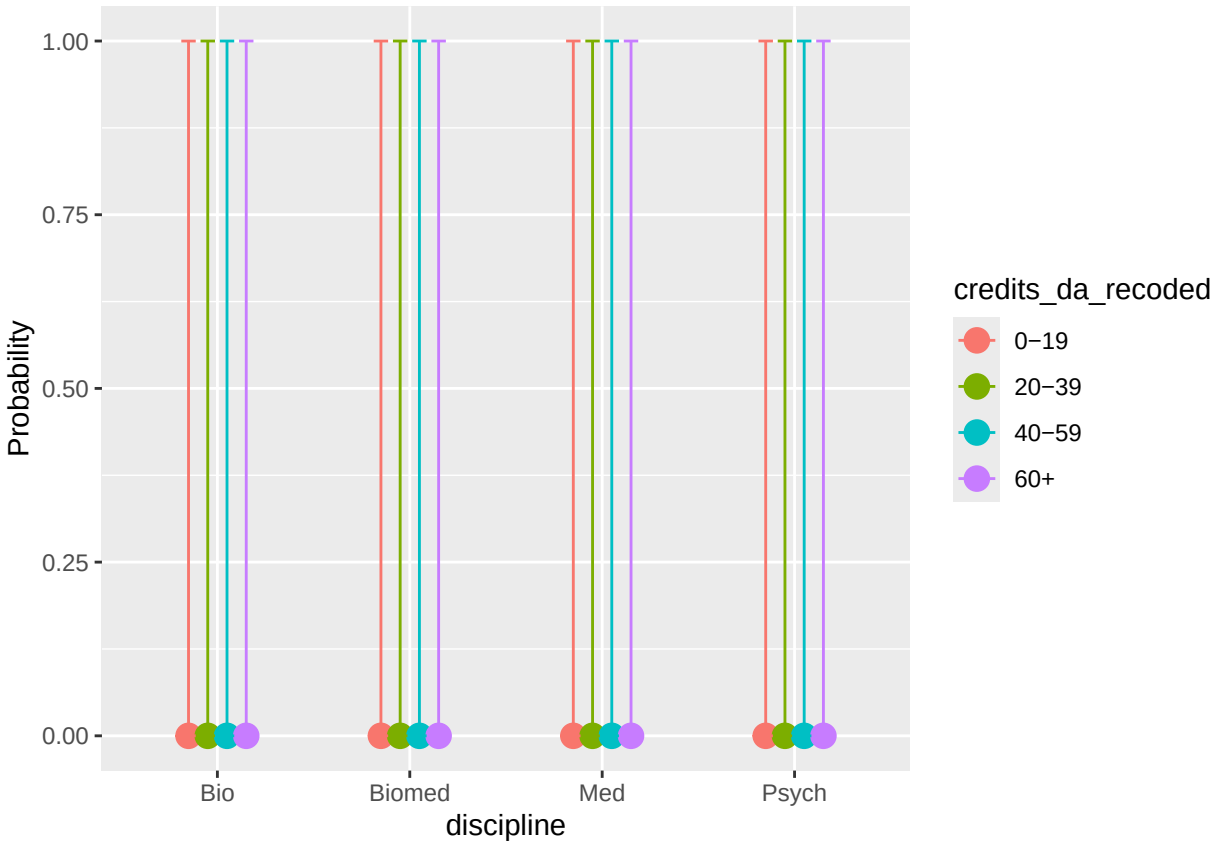
```
##  Family: categorical
##   Links: mu2039 = logit; mu4059 = logit; mu60 = logit
## Formula: credits_da_recoded ~ discipline
##    Data: survey_data (Number of observations: 103)
##   Draws: 4 chains, each with iter = 24000; warmup = 12000; thin = 1;
##          total post-warmup draws = 48000
##
## Regression Coefficients:
##                     Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## mu2039_Intercept     -35.88   2296.75  -161.25   181.52 1.00    14775     3424
## mu4059_Intercept     -18.93    972.49  -135.16   130.89 1.00    19892     5294
## mu60_Intercept         8.69    254.15  -114.07   132.76 1.00    25895     6096
## mu2039_discipline1     0.01     71.98   -32.00    35.29 1.00    15467     4054
## mu2039_discipline2     0.13     40.00   -31.08    30.94 1.00    24317     6142
## mu2039_discipline3     0.82     72.87   -31.09    33.74 1.00    21264     5521
## mu4059_discipline1    -0.42     71.54   -31.62    30.34 1.00    25165     6650
## mu4059_discipline2    -0.38     64.89   -31.83    29.48 1.00    15087     4920
## mu4059_discipline3    -0.79     65.72   -34.52    31.17 1.00    16688     4913
## mu60_discipline1      -0.34     49.15   -30.13    30.85 1.00    19908     5166
## mu60_discipline2       1.41     58.22   -27.39    31.24 1.00    18129     4413
## mu60_discipline3       1.00     32.23   -25.66    30.13 1.00    23684     5818
##
## Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
## and Tail_ESS are effective sample size measures, and Rhat is the potential
## scale reduction factor on split chains (at convergence, Rhat = 1).
```

```r
plot(brm.mn.dataanalysis.priorpc)
```


b_mu2039_Intercept
b_mu4059_Intercept
b_mu60_Intercept
b_mu2039_discipline1
b_mu2039_discipline2
b_mu2039_discipline3
b_mu4059_discipline1
b_mu4059_discipline2
b_mu4059_discipline3
b_mu60_discipline1
Chain
1
2
3
4




```
#Prior predictive check outout:
conditional_effects(brm.mn.dataanalysis.priorpc, categorical = TRUE)
```



Output from `summary(brm.mn.dataanalysis.priorpc)` and `plot(brm.mn.dataanalysis.priorpc)` both indicate that estimates are centered on zero (the prior has little influence on estimates).

NOTE: exploratory analyses (not shown) indicate that stronger priors yield estimates that approach 25% for each discipline (with narrower credible intervals) in prior predictive checks (`conditional_effects(brm.mn.dataanalysis.pri categorical = TRUE)`).

Now we run the multinomial model with our specified prior, and check for appropriate model behaviour:

```
#Run the model with the Gelman-ish prior:
brm.mn.dataanalysis <- brm(credits_da_recoded ~ discipline,
                      family = categorical(link = "logit"),
                      prior = priors.dataanalysis,
                      iter = 6000,
                      control = list(adapt_delta = 0.999, stepsize = 0.001,
                                     max_treedepth = 20),
                      data = survey_data,
                      file = here("models.gelman", "brm.mn.dataanalysis"))

#Examine prior that we specified:
prior_summary(brm.mn.dataanalysis)
```

```
##           prior     class        coef group resp   dpar nlpar lb ub tag
##  cauchy(0, 2.5)         b                        mu2039
##  cauchy(0, 2.5)         b disciplineBio          mu2039
##  cauchy(0, 2.5)         b disciplineBiomed       mu2039
##  cauchy(0, 2.5)         b disciplineMed          mu2039
##  cauchy(0, 2.5)         b disciplinePsych        mu2039
##  cauchy(0, 2.5)         b                        mu4059
##  cauchy(0, 2.5)         b disciplineBio          mu4059
```

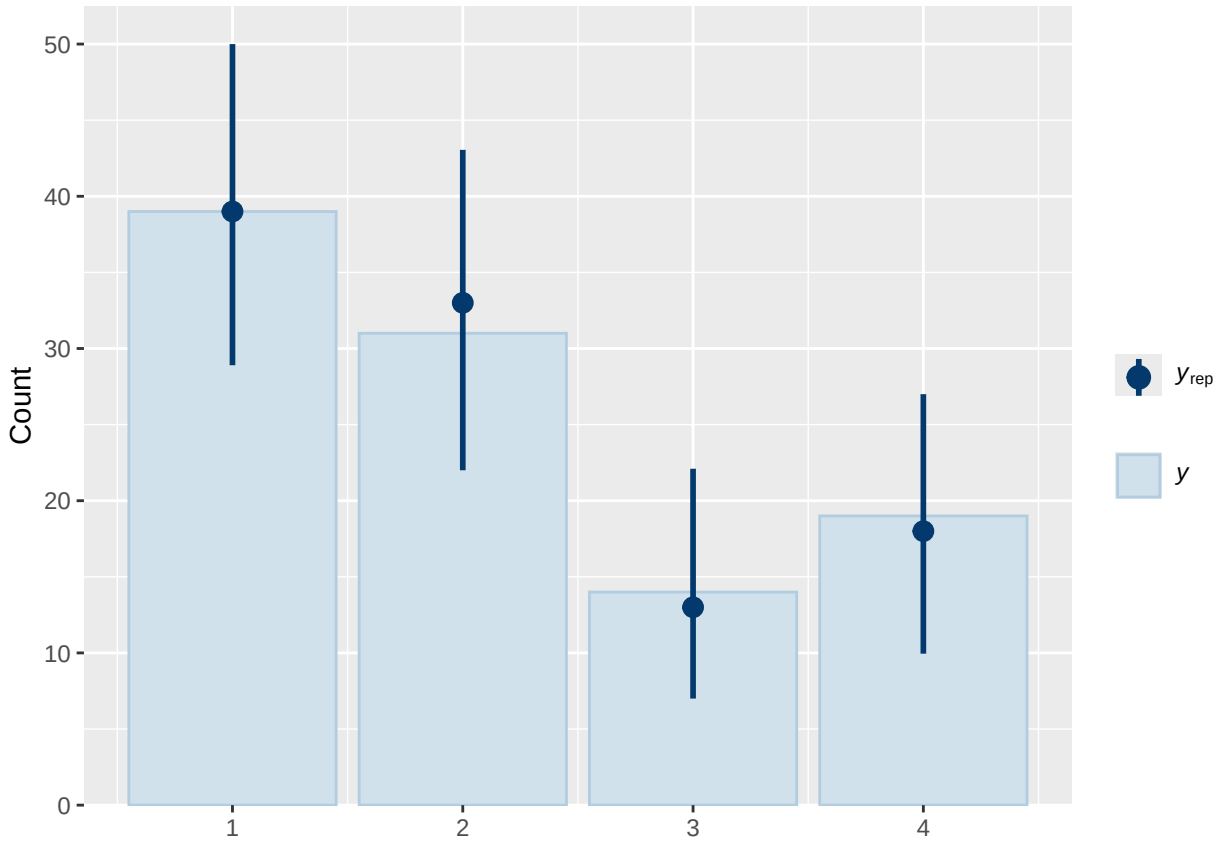
```
##  cauchy(0, 2.5)         b discipline2            mu4059
##  cauchy(0, 2.5)         b discipline3            mu4059
##  cauchy(0, 2.5)         b                          mu60
##  cauchy(0, 2.5)         b discipline1              mu60
##  cauchy(0, 2.5)         b discipline2              mu60
##  cauchy(0, 2.5)         b discipline3              mu60
##   cauchy(0, 10) Intercept                        mu2039
##   cauchy(0, 10) Intercept                        mu4059
##   cauchy(0, 10) Intercept                          mu60
##        source
##          user
##  (vectorized)
##  (vectorized)
##  (vectorized)
##          user
##  (vectorized)
##  (vectorized)
##  (vectorized)
##          user
##  (vectorized)
##  (vectorized)
##  (vectorized)
##          user
##          user
##          user
```

```
#Check convergence and coverage:
#We want Rhat to be less than 1.01
#We want _ESS to be 100 times the number of chains we run
summary(brm.mn.dataanalysis)
```

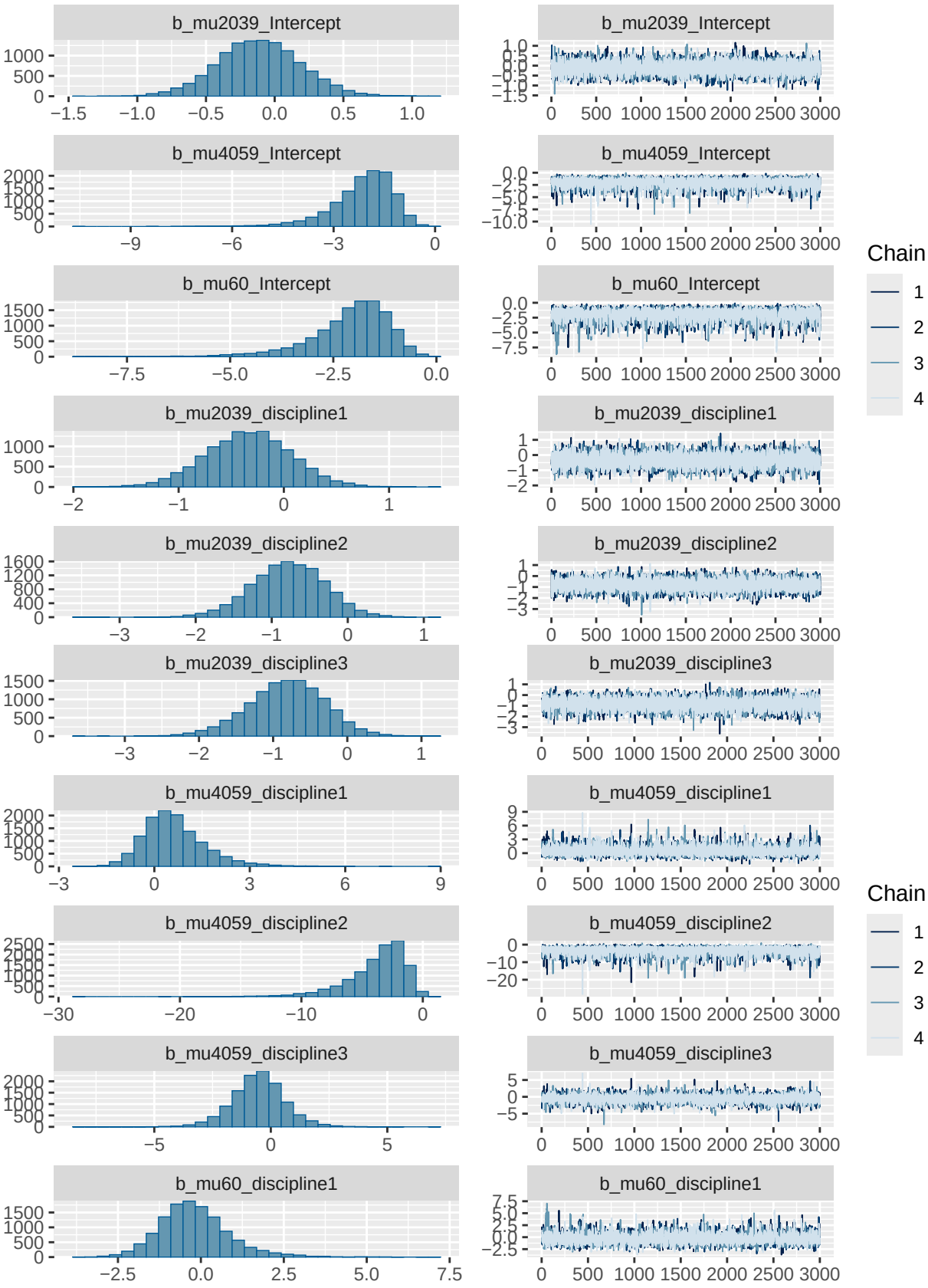
```
##  Family: categorical
##   Links: mu2039 = logit; mu4059 = logit; mu60 = logit
## Formula: credits_da_recoded ~ discipline
##    Data: survey_data (Number of observations: 103)
##   Draws: 4 chains, each with iter = 6000; warmup = 3000; thin = 1;
##          total post-warmup draws = 12000
##
## Regression Coefficients:
##                    Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## mu2039_Intercept      -0.11      0.31    -0.71     0.49 1.00     7411     7242
## mu4059_Intercept      -2.13      0.91    -4.33    -0.79 1.00     3848     3421
## mu60_Intercept        -2.09      0.97    -4.56    -0.71 1.00     3099     2527
## mu2039_discipline1    -0.34      0.41    -1.14     0.46 1.00     7683     7197
## mu2039_discipline2    -0.78      0.48    -1.75     0.13 1.00     7405     6743
## mu2039_discipline3    -0.82      0.52    -1.89     0.15 1.00     7134     7875
## mu4059_discipline1     0.63      0.98    -0.95     2.92 1.00     4124     3702
## mu4059_discipline2    -3.75      2.41    -9.86    -0.64 1.00     3580     3036
## mu4059_discipline3    -0.53      1.13    -2.81     1.72 1.00     4481     4416
## mu60_discipline1      -0.19      1.07    -2.04     2.29 1.00     3712     2726
## mu60_discipline2       0.06      1.08    -1.81     2.57 1.00     3539     2962
## mu60_discipline3      -4.02      2.69   -10.96    -0.58 1.00     3169     2373
##
## Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
## and Tail_ESS are effective sample size measures, and Rhat is the potential
```

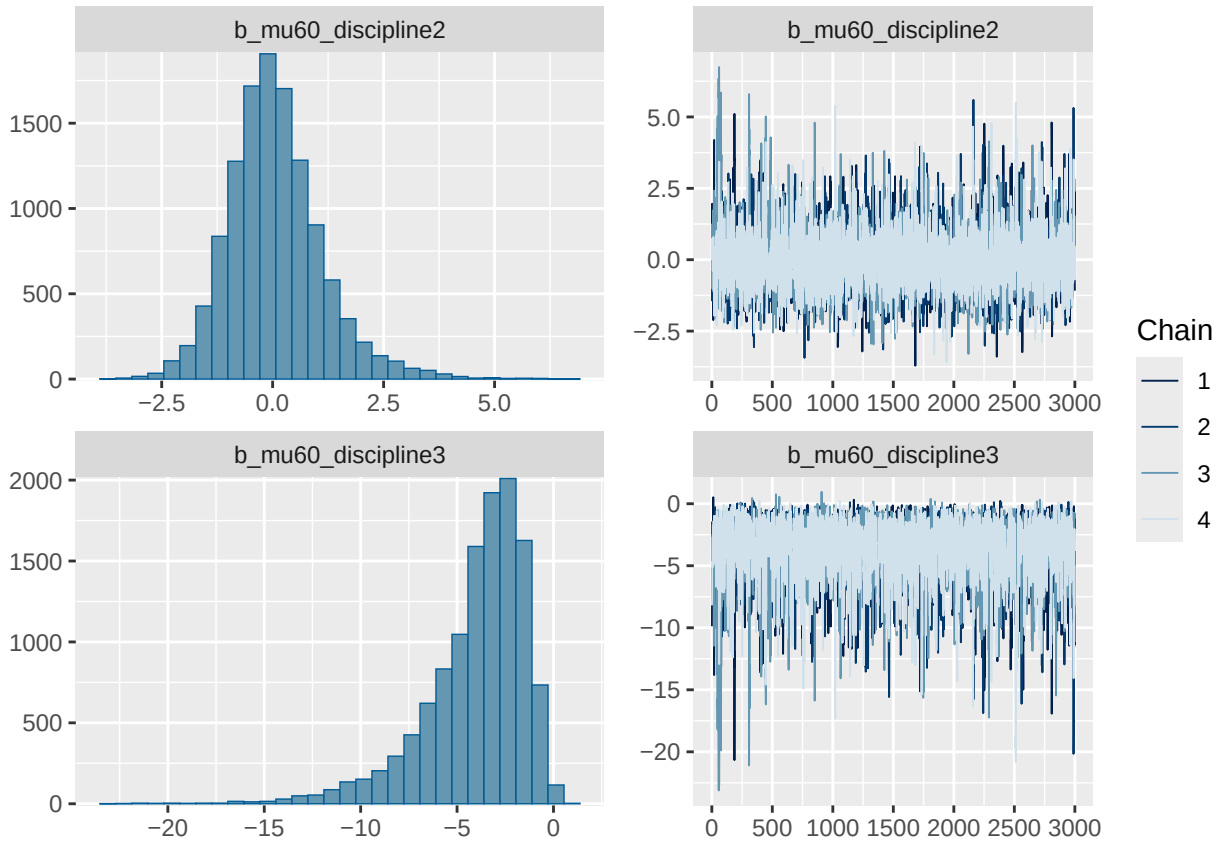
```
## scale reduction factor on split chains (at convergence, Rhat = 1).
```

```
#Posterior predictive check:
pp_check(brm.mn.dataanalysis, type = "bars", ndraws = 100)
```




```
#Check traces
plot(brm.mn.dataanalysis)
```


b_mu2039_Intercept
b_mu4059_Intercept
b_mu60_Intercept
b_mu2039_discipline1
b_mu2039_discipline2
b_mu2039_discipline3
b_mu4059_discipline1
b_mu4059_discipline2
b_mu4059_discipline3
b_mu60_discipline1
Chain
1
2
3
4



The summary (`summary(brm.mn.dataanalysis)`) indicates that the model converged well (based on Rhat) and has high coverage for all estimates.

The posterior predictive check output (`pp_check(brm.mn.dataanalysis, type = "bars", ndraws = 100)`) indicates sufficient model fit.

The traces and distributions of estimates on the latent scale (`plot(brm.mn.dataanalysis)`) also indicate good model behaviour.

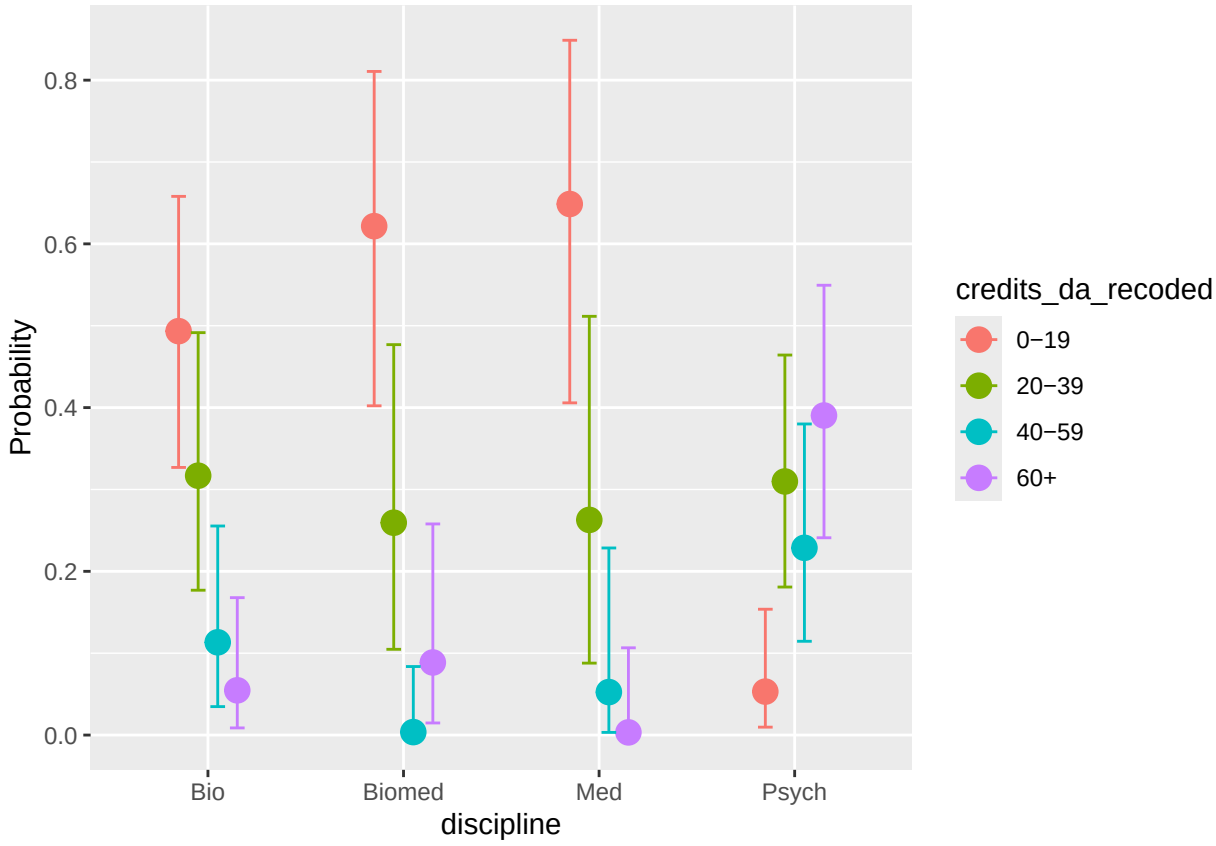
Now we obtain the model estimates:

```
eff_brm.mn.dataanalysis <- conditional_effects(brm.mn.dataanalysis, categorical = TRUE)
eff_brm.mn.dataanalysis[[1]]
```

```
##    discipline credits_da_recoded cond__ cats__ effect1__ effect2__  estimate__
## 1         Bio               0-19      1   0-19       Bio      0-19 0.493405164
## 2      Biomed               0-19      1   0-19    Biomed      0-19 0.621655907
## 3         Med               0-19      1   0-19       Med      0-19 0.648719960
## 4       Psych               0-19      1   0-19     Psych      0-19 0.053103312
## 5         Bio               0-19      1  20-39       Bio     20-39 0.316845233
## 6      Biomed               0-19      1  20-39    Biomed     20-39 0.259401423
## 7         Med               0-19      1  20-39       Med     20-39 0.262847164
## 8       Psych               0-19      1  20-39     Psych     20-39 0.309787871
## 9         Bio               0-19      1  40-59       Bio     40-59 0.113320109
## 10     Biomed               0-19      1  40-59    Biomed     40-59 0.003630282
## 11        Med               0-19      1  40-59       Med     40-59 0.052524841
## 12      Psych               0-19      1  40-59     Psych     40-59 0.228736286
## 13        Bio               0-19      1    60+       Bio       60+ 0.054650011
## 14     Biomed               0-19      1    60+    Biomed       60+ 0.088675714
```

```
## 15        Med               0-19      1    60+       Med       60+ 0.003279054
## 16      Psych               0-19      1    60+     Psych       60+ 0.390370571
##             se__      lower__    upper__
## 1   0.087631149 3.269112e-01 0.65801336
## 2   0.108700394 4.021917e-01 0.81064450
## 3   0.120457186 4.058183e-01 0.84872431
## 4   0.034329762 9.671284e-03 0.15372942
## 5   0.081216425 1.769623e-01 0.49157973
## 6   0.099778126 1.047750e-01 0.47693154
## 7   0.111195134 8.789712e-02 0.51151042
## 8   0.074733027 1.808149e-01 0.46420092
## 9   0.055465159 3.475375e-02 0.25538137
## 10  0.005335520 5.120923e-07 0.08372773
## 11  0.048478897 3.251230e-03 0.22854383
## 12  0.068472758 1.145772e-01 0.38000757
## 13  0.037764664 8.741620e-03 0.16784482
## 14  0.059792629 1.478278e-02 0.25790428
## 15  0.004838595 1.286873e-07 0.10658689
## 16  0.079399008 2.410207e-01 0.54944819
```

```
print(eff_brm.mn.dataanalysis)
```



Plot

```
data_analysis_estimates <- eff_brm.mn.dataanalysis[[1]] %>%
  clean_names() %>%
```

```
  rename(
    credits_da = cats,
    probability_estimate = estimate) %>%
  select(discipline, credits_da, probability_estimate, lower, upper)

data_analysis_estimates
```

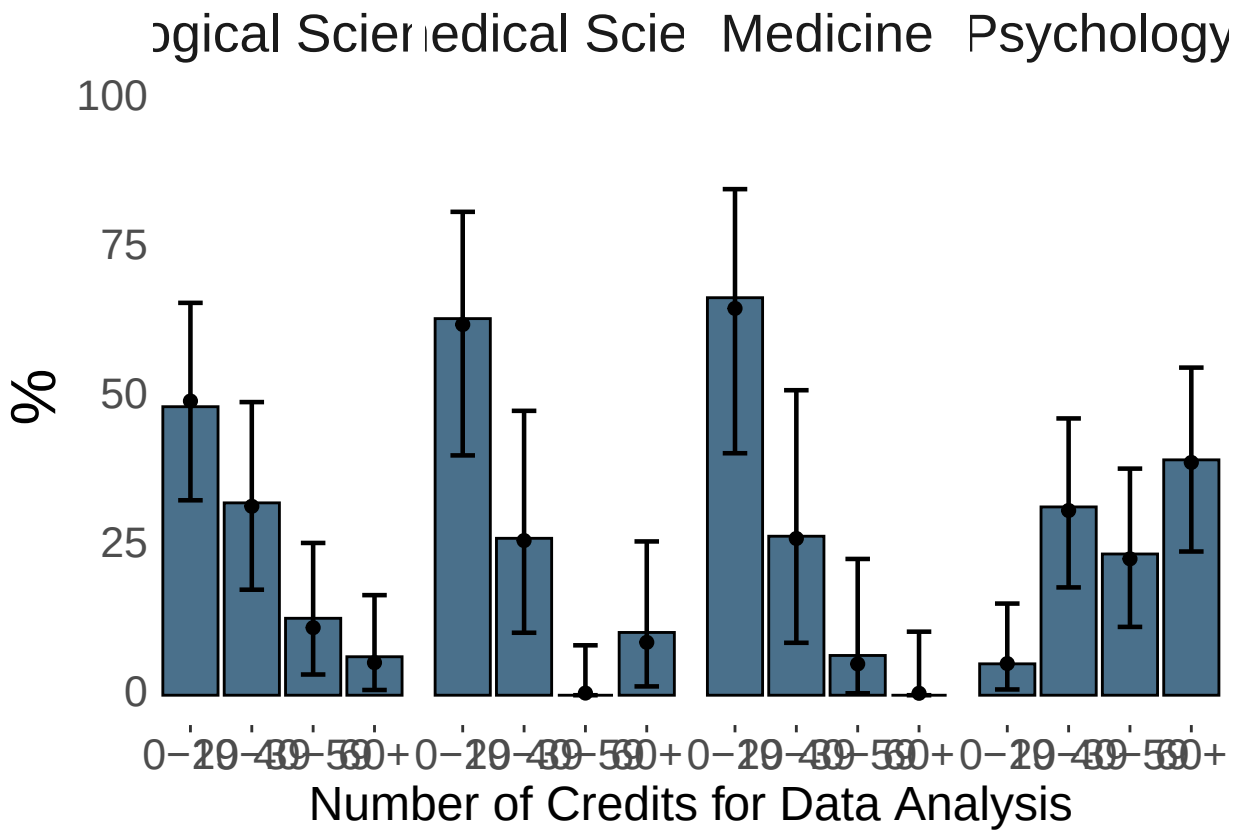
```
##    discipline credits_da probability_estimate        lower      upper
## 1         Bio       0-19          0.493405164 3.269112e-01 0.65801336
## 2      Biomed       0-19          0.621655907 4.021917e-01 0.81064450
## 3         Med       0-19          0.648719960 4.058183e-01 0.84872431
## 4       Psych       0-19          0.053103312 9.671284e-03 0.15372942
## 5         Bio      20-39          0.316845233 1.769623e-01 0.49157973
## 6      Biomed      20-39          0.259401423 1.047750e-01 0.47693154
## 7         Med      20-39          0.262847164 8.789712e-02 0.51151042
## 8       Psych      20-39          0.309787871 1.808149e-01 0.46420092
## 9         Bio      40-59          0.113320109 3.475375e-02 0.25538137
## 10     Biomed      40-59          0.003630282 5.120923e-07 0.08372773
## 11        Med      40-59          0.052524841 3.251230e-03 0.22854383
## 12      Psych      40-59          0.228736286 1.145772e-01 0.38000757
## 13        Bio        60+          0.054650011 8.741620e-03 0.16784482
## 14     Biomed        60+          0.088675714 1.478278e-02 0.25790428
## 15        Med        60+          0.003279054 1.286873e-07 0.10658689
## 16      Psych        60+          0.390370571 2.410207e-01 0.54944819
```

```
survey_table_da <- table(survey_data$credits_da_recoded, survey_data$discipline)
survey_table_da <- prop.table(survey_table_da, margin = 2)
survey_table_da <- as.data.frame(survey_table_da)
survey_ggplot_da <- survey_table_da %>%
  rename(
    credits_da = Var1,
    discipline = Var2) %>%
  mutate(percentage = 100*Freq)
```

```
data_analysis_plot <- ggplot() +
  geom_col(
    data = survey_ggplot_da,
    aes(x = credits_da, y = percentage),
    color = "black",
    fill = plot_colour) +
  geom_errorbar(
    data = data_analysis_estimates,
    aes(x = credits_da, ymin = 100*lower, ymax = 100*upper),
    width = .4,
    linewidth = .8) +
  geom_point(
    data = data_analysis_estimates,
    aes(x = credits_da, y = 100*probability_estimate),
    size = 2) +
  scale_x_discrete(drop = FALSE) +
  scale_y_continuous(
    "%",
    limits = c(0, 100),
    labels = c(0, 25, 50, 75, 100)) +
```

```
  facet_wrap(
    ~ factor(
      discipline,
      labels = c("Biological Sciences", "Biomedical Sciences", "Medicine", "Psychology")),
    nrow = 1) +
  xlab("Number of Credits for Data Analysis") +
  plot_theme


data_analysis_plot
```




```
ggsave(
  plot = data_analysis_plot,
  here("figures", "credits_da_plot_gelman.jpg"),
  device = "jpeg",
  width = 1280,
  height = 637,
  units = "px",
  dpi = 100)
```

Perform Sensitivity Analysis:

```
#Sensitivity analysis:  determines the extent to which the prior affects the results
#(a qualitative test, not a quantitative test)

brm.mn_fixed <- tidy(brm.mn.dataanalysis, effects = "fixed", conf.level = 0.95, fix.intercept = FALSE) %
  mutate(
    ci_width = abs(conf.low - conf.high)
```

```
  )
```

```
## Warning in tidy.brmsfit(brm.mn.dataanalysis, effects = "fixed", conf.level =
## 0.95, : some parameter names contain underscores: term naming may be
## unreliable!
```

```
brm.mn_fixed
```

```
## # A tibble: 12 x 8
##    effect component term          estimate std.error conf.low conf.high ci_width
##    <chr>  <chr>     <chr>            <dbl>     <dbl>    <dbl>     <dbl>    <dbl>
##  1 fixed  cond      mu2039_Inter~  -0.111      0.305   -0.708     0.491     1.20
##  2 fixed  cond      mu4059_Inter~  -2.13       0.907   -4.33     -0.787     3.55
##  3 fixed  cond      mu60_Interce~  -2.09       0.971   -4.56     -0.706     3.86
##  4 fixed  cond      mu2039_disci~  -0.336      0.407   -1.14      0.456     1.59
##  5 fixed  cond      mu2039_disci~  -0.782      0.482   -1.75      0.130     1.88
##  6 fixed  cond      mu2039_disci~  -0.820      0.515   -1.89      0.146     2.04
##  7 fixed  cond      mu4059_disci~   0.629      0.975   -0.951     2.92      3.87
##  8 fixed  cond      mu4059_disci~  -3.75       2.41    -9.86     -0.635     9.23
##  9 fixed  cond      mu4059_disci~  -0.529      1.13    -2.81      1.72      4.54
## 10 fixed  cond      mu60_discipl~  -0.189      1.07    -2.04      2.29      4.32
## 11 fixed  cond      mu60_discipl~   0.0629     1.08    -1.81      2.57      4.38
## 12 fixed  cond      mu60_discipl~  -4.02       2.69   -11.0      -0.576    10.4
```

```
brm.mn_fixed_reduced <- brm.mn_fixed %>%
  select(
    effect, component, term, estimate
  )
brm.mn_fixed_reduced
```

```
## # A tibble: 12 x 4
##    effect component term                estimate
##    <chr>  <chr>     <chr>                  <dbl>
##  1 fixed  cond      mu2039_Intercept     -0.111
##  2 fixed  cond      mu4059_Intercept     -2.13
##  3 fixed  cond      mu60_Intercept       -2.09
##  4 fixed  cond      mu2039_discipline1   -0.336
##  5 fixed  cond      mu2039_discipline2   -0.782
##  6 fixed  cond      mu2039_discipline3   -0.820
##  7 fixed  cond      mu4059_discipline1    0.629
##  8 fixed  cond      mu4059_discipline2   -3.75
##  9 fixed  cond      mu4059_discipline3   -0.529
## 10 fixed  cond      mu60_discipline1     -0.189
## 11 fixed  cond      mu60_discipline2      0.0629
## 12 fixed  cond      mu60_discipline3     -4.02
```

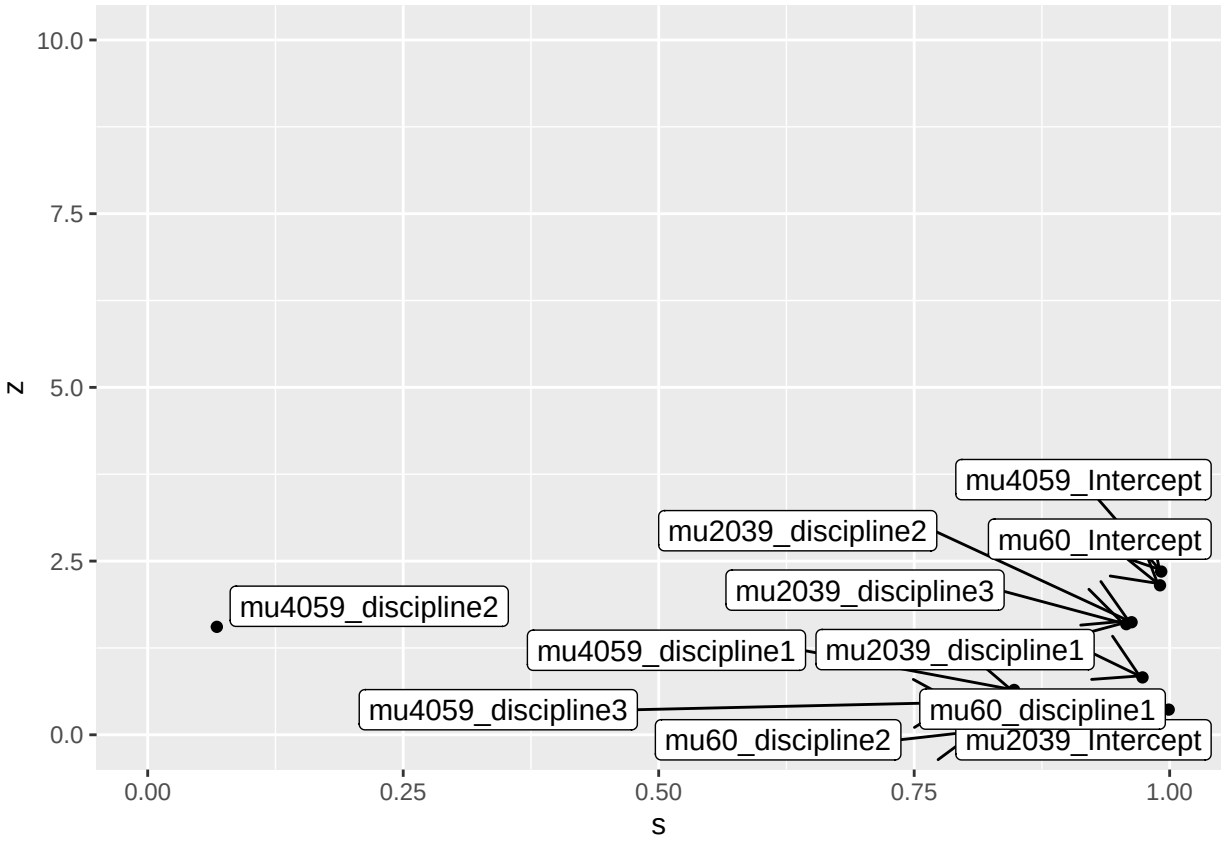
```
#We have 12 fixed effects; the first 3 are intercepts, and the last 9 are coefficients (b)
#Also, the 3 Intercepts are listed first in the output of brm.mn_fixed,
#and the Intercepts had priors with sd = 10
#The remaining 9 fixed effects are for b, which had sd = 2.5 for the prior.
#Hence, we set up the sensitivity analysis like this:
brm.mn.fixed <- brm.mn_fixed %>%
  mutate(
    theta = c(0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0),
    sigma_prior = c(10, 10, 10, 2.5, 2.5, 2.5, 2.5, 2.5, 2.5, 2.5, 2.5, 2.5),
    z = abs((estimate - theta) / std.error), # it's called here std.error but is the standard deviation
```

```
    s = 1 - (std.error^2 / sigma_prior^2)
  )

  ggplot(brm.mn.fixed, aes(s, z, label = term)) +
  geom_point() +
  geom_label_repel(arrow = arrow()) +
  xlim(0, 1) + ylim(0, 10)
```

```
## Warning: Removed 1 row containing missing values or values outside the scale range
## (`geom_point()`).
```

```
## Warning: Removed 1 row containing missing values or values outside the scale range
## (`geom_label_repel()`).
```




```
  brm.mn.fixed$s
```

```
##   b_mu2039_Intercept   b_mu4059_Intercept     b_mu60_Intercept
##           0.99906888           0.99178024           0.99057412
## b_mu2039_discipline1 b_mu2039_discipline2 b_mu2039_discipline3
##           0.97345768           0.96276345           0.95756393
## b_mu4059_discipline1 b_mu4059_discipline2 b_mu4059_discipline3
##           0.84782385           0.06764366           0.79401325
##   b_mu60_discipline1   b_mu60_discipline2   b_mu60_discipline3
##           0.81594727           0.81475738          -0.15675127
```

```
  brm.mn.fixed$z
```

```
##   b_mu2039_Intercept   b_mu4059_Intercept     b_mu60_Intercept
##           0.36254913           2.34958547           2.15139834
```

```
## b_mu2039_discipline1 b_mu2039_discipline2 b_mu2039_discipline3
##           0.82602527           1.62032940           1.59234879
## b_mu4059_discipline1 b_mu4059_discipline2 b_mu4059_discipline3
##           0.64544456           1.55388973           0.46665759
##   b_mu60_discipline1   b_mu60_discipline2   b_mu60_discipline3
##           0.17584807           0.05848888           1.49567278
```